3. Motion in a Plane

Can you recall?

1. What is meant by motion? 2. What is rectilinear motion?
3. What is the difference between displacement and distance travelled?
4. What is the difference between uniform and nonuniform motion?

3.1 Introduction:

We see objects moving all around us. Motion is a change in the position of an object with time. We have come across the motion of a toy car when pushed along some particular direction, the motion of a cricket ball hit by a batsman for a sixer and the motion of an aeroplane from one place to another. The motion of objects can be divided in three categories: (1) motion along a straight line, i.e., rectilinear motion, (2) motion in two dimensions, i.e., motion in a plane and, (3) motion in three dimensions, i.e., motion in space. The above cited examples correspond to three types of motions, respectively. You have studied rectilinear motion in earlier standards. In rectilinear motion the force acting on the object and the velocity of the object both are along one and the same line. The distances are measured along the line only and we can indicate distances along the +ve and –ve axes as being positive and negative, respectively. The study of the motion of an object in a plane or in space becomes much easier and the corresponding equations become more elegant if we use vector quantities. In this Chapter we will first recall basic facts about rectilinear motion. We will use vector notation for this study as it will be useful later when we will study the motion in two dimensions. We will then study the motion in two dimensions which will be restricted to projectile motion only. Circular motion, i.e., the motion of an object around a circular path will be introduced here and will be studied in detail in the next standard.

3.2 Rectilinear Motion:

Consider an object moving along a straight line. Let us assume this line to be along the x-axis. Let $\vec{x}_1$ and $\vec{x}_2$ be the position vectors of the body at times t_1 and t_2 during its motion. The following quantities can be defined for the motion.

1. **Displacement:** The displacement of the object between t_1 and t_2 is the difference between the position vectors of the object at the two instances. Thus, the displacement is given by

$$\vec{s} = \Delta\vec{x} = \vec{x}_2 - \vec{x}_1 \qquad \text{--- (3.1)}$$

Its direction is along the line of motion of the object. Its dimensions are that of length. For example, if an object has travelled through 1 m from time t_1 to t_2 along the +ve x-direction, the magnitude of its displacement is 1 m and its direction is along the +ve x-axis. On the other hand, if the object travelled along the +ve y direction through the same distance in the same time, the magnitude of its displacement is the same as before, i.e., 1 m but the direction of the displacement is along the +ve y-axis.

2. **Path length**: This is the actual distance travelled by the object during its motion. It is a scalar quantity and its dimensions are also that of length. If an object travels along the x-axis from x = 2 m to x = 5 m then the distance travelled is 3 m. In this case the displacement is also 3 m and its direction is along the +ve x-axis. However, if the object now comes back to x = 4, then the distance through which the object has moved increases to 3 + 1 = 4 m. Its initial position was x = 2 m and the final position is now x = 4 m and thus, its displacement is Δx = 4 – 2 = 2 m, i.e., the magnitude of the displacement is 2 m and its direction is along the +ve x-axis. If the object now moves to x =1, then the distance travelled, i.e., the path length increases to 4 + 3 =

7 m while the magnitude of displacement becomes 2 – 1 = 1 m and its direction is along the negative x-axis.

3. **Average velocity**: This is defined as the displacement of the object during the time interval over which average velocity is being calculated, divided by that time interval. As displacement is a vector quantity, the velocity is also a vector quantity. Its dimensions are $[L^1 M^0 T^{-1}]$. If the position vectors of the object are $\vec{x}_1$ and $\vec{x}_2$ at times t_1 and t_2 respectively, then the average velocity is given by

$$\vec{v}_{av} = \frac{\vec{x}_2 - \vec{x}_1}{(t_2 - t_1)} \qquad \text{--- (3.2)}$$

For example, if the positions of an object are $x = +2$ m and $x = +4$ m at times $t = 0$ and $t = 1$ minute respectively, the magnitude of its average velocity during that time is $v_{av} = (4 - 2)/(1 - 0) = 2$ m per minute and its direction will be along the +ve x-axis, and we write $\vec{v}_{av} = 2\hat{i}$ m/min where $\hat{i}$ is a unit vector along x-axis.

4. **Average speed**: This is defined as the total path length travelled during the time interval over which average speed is being calculated, divided by that time interval.

Average speed = v_{av} = path length/time interval. It is a scalar quantity and has the same dimensions as that of velocity, i.e., $[L^1 M^0 T^{-1}]$.

If the rectilinear motion of the object is only in one direction along a line, then the magnitude of its displacement will be equal to the distance travelled and so the magnitude of average velocity will be equal to the average speed. However if the object reverses its direction (the motion remaining along the same line) then the magnitude of displacement will be smaller than the path length and the average speed will be larger than the magnitude of average velocity.

5. **Instantaneous velocity**: Instantaneous velocity of an object is its velocity at a given instant of time. It is defined as the limiting value of the average velocity of the object over a small time interval (Δt) around t when the value of the time interval (Δt) goes to zero.

$$\vec{v} = \lim_{\Delta t \to 0}\left(\frac{\Delta \vec{x}}{\Delta t}\right) = \frac{d\vec{x}}{dt}, \qquad \text{--- (3.3)}$$

$\frac{d\vec{x}}{dt}$ being the derivative of $\vec{x}$ with respect to t (see Chapter 2).

6. **Instantaneous speed:** Instantaneous speed is the speed of an object at a given instant of time t. It is the limiting value of the average speed of the object taken over a small time interval (Δt) around t when the time interval goes to zero. In such a limit, the path length will be equal to the magnitude of the displacement and so the instantaneous speed will always be equal to the magnitude of the instantaneous velocity of the object.

Always Remember:

For uniform rectilinear motion, i.e., for an object moving with constant velocity along a straight line

1. The average and instantaneous velocities are equal.
2. The average and instantaneous speeds are the same and are equal to the magnitude of the velocity.

For nonuniform rectilinear motion

1. The average and instantaneous velocities are different.
2. The average and instantaneous speeds are different.
3. The average speed will be different from the magnitude of average velocity.

Example 3.1: A person walks from point P to point Q along a straight road in 10 minutes, then turns back and returns to point R which is midway between P and Q after further 4 minutes. If PQ is 1 km, find the average speed

and velocity of the person in going from P to R.

Solution: The path length travelled by the person is 1.5 km while the displacement is the distance between R and P which is 0.5 km. The time taken for the motion is 14 min.

The average speed = 1.5 / 14 = 0.107 km/min = 6.42 km/hr.

The magnitude of the average velocity = 0.5/14 = 0.0357 km/min = 2.142 km/hr.

Graphical Study of Motion

We can study the motion of an object by using graphs showing its position as a function of time. Figure 3.1 shows the graphs of position as a function of time for five different types of motion of an object. Figure 3.1(a) shows an object at rest, for which the *x*-*t* graph is a horizontal straight line. Since the position is not changing, displacement of the object zero. Velocity is displacement (which is zero) divided by time interval or the derivative of displacement with respect to time. It can be obtained from the slope of the line plotted in the figure which is zero.

Figure 3.1(b) shows *x*-*t* graph for an object moving with constant velocity along the +ve *x*-axis. Since velocity is constant, displacement is proportional to elapsed time. The slope of the straight line is +ve, showing that the velocity is along the +ve *x*-axis. As the motion is uniform, the average velocity is same as the instantaneous velocity at all times. Also, the speed is equal to the magnitude of the velocity.

Figure 3.1(c) shows the *x*-*t* graph for a body moving with uniform velocity but along the -ve *x*-axis, the slope of the line being -ve. Figure 3.1(d) shows the *x*-*t* graph of an object having oscillatory motion with constant speed. The direction of velocity changes from +ve to -ve and vice versa over fixed intervals of time.

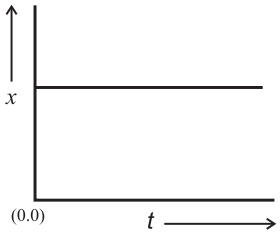


Fig 3.1 (a): Object at rest.

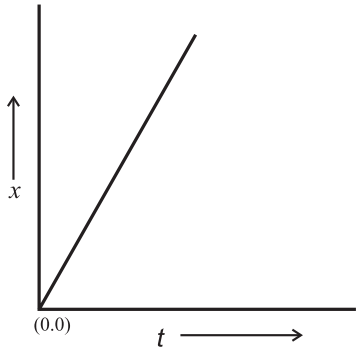


Fig 3.1 (b): Object with uniform velocity along +ve *x*-axis.

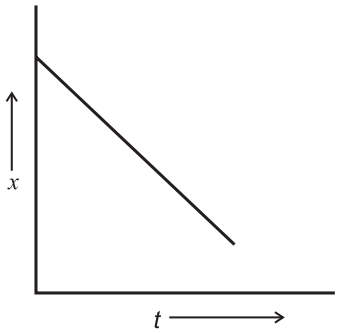


Fig 3.1 (c): Object with uniform velocity along -ve *x*-axis.

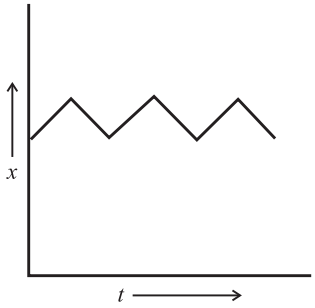


Fig 3.1 (d): Object performing oscillatory motion.

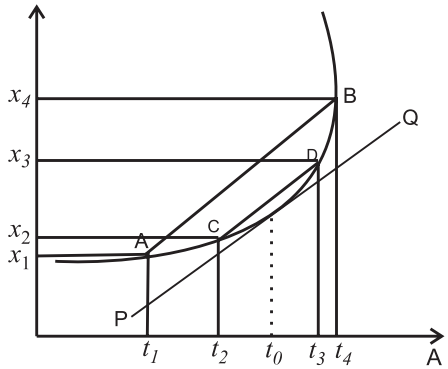


Fig.3.1 (e): Object in nonuniform motion.

Figure 3.1(e) shows the motion of an

object with nonuniform velocity. Its velocity changes with time and, therefore, the average and instantaneous velocities are different. Figure shows the average velocity over time interval from t_1 to t_4 around time t_0, which can be seen from Eq. (3.2) to be the slope of line AB. For a smaller time interval from t_2 to t_3, the average velocity is the slope of the line CD. If we keep reducing the time interval around t_0, we will ultimately come to a limit, when the time interval will go to zero and lines AB, CD... will go over to the tangent to the curve at t_0. The instantaneous velocity at t_0 will thus be equal to the slope of the tangent PQ at t_0 (see Eq. (3.3)).

7. Acceleration: Acceleration is defined as the rate of change of velocity with time. It is a vector quantity and its dimensions are $[L^1 M^0 T^{-2}]$. The average acceleration of an object having velocities $\vec{v}_1$ and $\vec{v}_2$ at times t_1 and t_2 is given by

$$\vec{a} = \frac{(\vec{v}_2 - \vec{v}_1)}{(t_2 - t_1)} \qquad \text{--- (3.4)}$$

Instantaneous acceleration is the limiting value of the average acceleration when the time interval goes to zero. It is given by

$$\vec{a} = \lim_{\Delta t \to 0}\left(\frac{\Delta \vec{v}}{\Delta t}\right) = \frac{d\vec{v}}{dt} \qquad \text{--- (3.5)}$$

The instantaneous acceleration at a given time is the slope of the tangent to the velocity versus time curve at that time. Figure 3.2 shows the velocity versus time (v - t) graphs for four different cases. Figure 3.2(a) represents the motion of an object with zero acceleration, i.e., constant velocity. The shaded area under the velocity-time graph over some time interval t_1 to t_2, shown in Figs. 3.2(a) is equal to v_0 $(t_2 - t_1)$ which is the magnitude of the displacement of the object from t_1 to t_2. Figure 3.2(b) is the velocity-time graph for an object moving with constant +ve acceleration (magnitude of velocity uniformly increasing with time). Figure 3.2(c) shows similar motion but the object has -ve acceleration, i.e., the acceleration is opposite to the direction of velocity which, therefore, decreases uniformly with time. The area under both the curves between two instants of time is the displacement of the object during that time interval (as shown below). Figure 3.2(d) shows the motion of an object having nonuniform acceleration. The average acceleration between t_1 and t_2 around t_0 and the instantaneous accelerations at t_0 for the object are shown by straight lines AB and CD respectively.

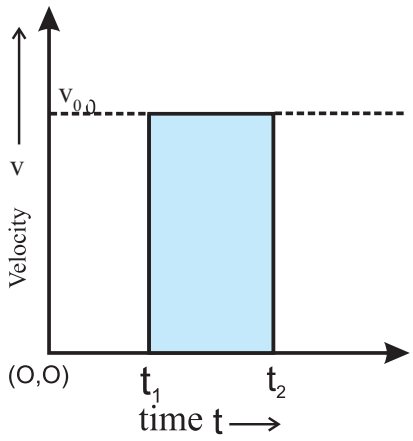


Fig 3.2 (a): Object moving with constant velocity.

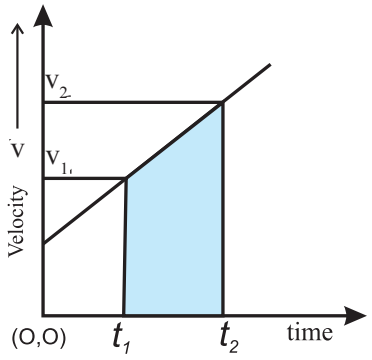


Fig 3.2 (b): Object moving with velocity (v) along +ve x-axis with uniform acceleration along the same direction.

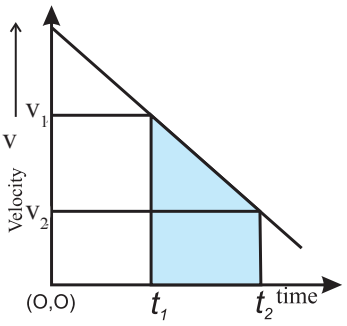


Fig 3.2 (c): Object moving with velocity (v) with negative uniform acceleration.

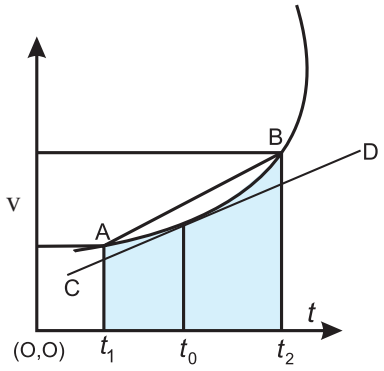


Fig. 3.2 (d): Object moving with nonuniform acceleration.

The area under the velocity-time curves in Figs. 3.2(a) to (d) can be written using the definition of integral given in Chapter 2 as

$$\text{Area} = \int_{t_1}^{t_2} v\,dt = \int_{t_1}^{t_2} \frac{dx}{dt}\,dt = \int_{t_1}^{t_2} dx = x(t_2) - x(t_1) \quad \text{--- (3.6)}$$

= displacement of the object from t_1 to t_2.

Always Remember:

For uniform acceleration, for a rectilinear motion:

1. Velocity-time graph is linear.
2. The area under the velocity-time graph between two instants of time t_1 and t_2 gives the displacement of the object during that time interval.
3. The slope of the velocity-time graph is the acceleration of the object

For nonuniform acceleration in a rectilinear motion:

1. Velocity-time graph is nonlinear.
2. The area under the velocity-time graph between two instants of time t_1 and t_2 gives the displacement of the object during that time interval.
3. The instantaneous acceleration of the object at a given time is equal to the slope of the tangent to the curve at that point.

While using the concept of area under the curve, the origin of the velocity axis (for v-t graph) must be zero.

Equations of Motion for Uniform Acceleration:

We can graphically derive Newton's equations of motion for an object moving with uniform acceleration. Consider an object having position $x = 0$ at $t = 0$. Let the velocity at $t = 0$ be u and at time t be v. The graphical representation of motion is shown in Fig. 3.3. The acceleration is given by the slope of the line AB. Thus,

$$\text{Acceleration, } a = \frac{v - u}{t - 0} = \frac{v - u}{t}$$

$$\therefore v = u + at \quad \text{--- (3.7)}$$

This is the first equation of motion.

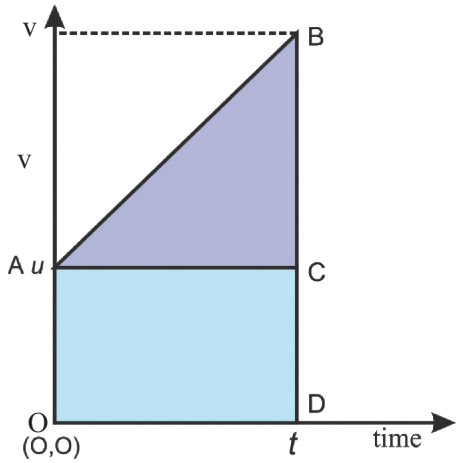


Fig.3.3: Derivation of equation of motion for motion with uniform acceleration.

As we know, the area under the curve in velocity-time graph is the displacement of the object. Thus displacement s = area of the quadrilateral OABD. = area of triangle ABC + area of rectangle OACD.

$$= \frac{1}{2}(v - u)t + ut$$

$$\text{Using Eq. (3.7), } s = ut + \frac{1}{2}at^2 \quad \text{--- (3.8)}$$

This is the second equation of motion.

As the acceleration is constant, the velocity is increasing linearly with time and we can use average velocity v_{av}, to calculate the displacement using Eq. (3.7) as

$$s = v_{av}t = \left(\frac{v + u}{2}\right)t = \frac{(v + u)(v - u)}{2a}$$

$$\therefore s = \left(v^2 - u^2\right)/(2a)$$

$$\therefore v^2 - u^2 = 2as \quad \text{--- (3.9)}$$

This is the third equation of motion. Vector notation was not included here as the motion was rectilinear.

The most common example of uniform rectilinear motion with uniform acceleration of an object in day to day life is a freely falling body. When a body starts with zero velocity at a certain height from the ground and falls under the influence of the gravity of the Earth , it is said to be in free fall. The only other force that acts on it is that of the air resistance or friction. For displacements of a few metres, this force is too small and can be neglected. The acceleration of the body is the acceleration due to gravity which is along the vertical direction and can be assumed to be constant over distances which are small compared to the radius of the Earth . Thus the velocity and acceleration are both along the vertical direction and the motion is a uniform rectilinear motion with uniform acceleration.



Do you know ?

The distance travelled by an object starting from rest and having a uniform acceleration in successive seconds are in the ratio 1:3:5:7... Consider a freely falling object. Let us calculate the distances travelled by it in equal intervals of time t_0 (say). This can be done using the second equation of motion s = $u\ t_0$ +(1/2) $g\ t_0^2$. The initial velocity is zero. Therefore, the distance travelled in the first t_0 interval = (1/2) $g\ t_0^2$. For simplification let us write (1/2) g = A. Then the distance travelled in the first t_0 time interval = $d_1 = At_0^2$. In the time interval $2t_0$, the distance travelled = $A(2t_0)^2$. Hence, the distance travelled in the second t_0 interval is $d_2 = A(4t_0^2 - t_0^2) = 3A\ t_0^2 = 3\ d_1$. The distance travelled in time interval $3t_0 = A(3t_0)^2$. Thus, the distance travelled in the 3rd t_0 interval = $d_3 = A(9t_0^2 - 4t_0^2) = 5A\ t_0^2 = 5d_1$. Continuing, one can see that the distances d_1, d_2, d_3 .. are in the ratio 1:3:5:7... This is true for any rectilinear motion, starting from rest, with positive uniform acceleration.

Example 3.2: A stone is thrown vertically upwards from the ground with a velocity 15 m/s. At the same instant a ball is dropped from a point directly above the stone from a height of 30 m. At what height from the ground will the stone and the ball meet and after how much time? (Use $g = 10\ m/s^2$ for ease of calculation).

Solution: Let us assume that the stone and the ball meet after time t_0. The distances (not displacements) travelled by the stone and the ball in that time can be obtained from Eq. (3.8) as

$$s_{stone} = 15\ t_0 - \frac{1}{2}\ g\ t_0^2$$

$$s_{ball} = \frac{1}{2}\ g\ t_0^2$$

When they meet, $s_{stone} + s_{ball} = 30$

$$15\ t_0 - \frac{1}{2}\ g\ t_0^2 + \frac{1}{2}\ g\ t_0^2 = 30$$

$$t_0 = 30/15 = 2\ s$$

$$\therefore\ s_{stone} = 15\ (2) - \frac{1}{2}\ (10)\ (2)^2 = 30 - 20 = 10\ m$$

Thus the stone and the ball meet at a height of 10 m.

8. Relative Velocity: You must have often experienced relative motion. The most striking example is when you are going in a train and another train travelling in the same direction along parallel tracks, overtakes you. If you look at that train, it actually seems to be moving much slower than what your train seemed to move and yet it is overtaking you. On the other hand if your train overtakes another train, travelling on a parallel track in the same direction, and you look at that train, you feel that your train has suddenly slowed down. Why does this happen? This is because when you look at the neighbouring train, you are actually experiencing relative motion, i.e., your motion with respect to the other train or the motion of the other train with respect to you. Thus, in the first case as the other train overtakes you what you perceive is the velocity of the train with respect to you, i.e., the difference in the velocities of the two trains which most often is much smaller than the velocity of your train. In the second case, you are moving faster but when you look at that train you only feel your velocity

relative to it and, therefore, your velocity appears to be lower than its actual value. We can define relative velocity of object A with respect to object B as the difference between their velocities, i.e.,

$$v_{AB} = v_A - v_B \quad \text{--- (3.10)}$$

Similarly, the velocity of B with respect to A is given by

$$v_{BA} = v_B - v_A \quad \text{--- (3.11)}$$

We assume that at time $t = 0$, A and B were at the same point $x = 0$. As they are travelling with different velocities, the distance between them will go on increasing with time in direct proportion to the difference in their velocities, i.e., the relative velocity between them.

Example 3.3: An aeroplane A, is travelling in a straight line with a velocity of 300 km/hr with respect to Earth. Another aeroplane B, is travelling in the opposite direction with a velocity of 350 km/hr with respect to Earth. What is the relative velocity of A with respect to B? What should be the velocity of a third aeroplane C moving parallel to A, relative to the Earth if it has a relative velocity of 100 km/hr with respect to A?

Solution: Let v_A, v_B and v_C be the velocities of the three planes relative to the Earth. Relative velocity of A with respect to B = $v_{AB} = v_A - v_B$ = 300 – (-350) = 650 km/hr

Relative velocity of C with respect to A = $v_{CA} = v_C - v_A$ = 100 km/hr.

Thus, $v_C = v_{CA} + v_A$ = 400 km/hr

3.3 Motion in Two Dimensions-Motion in a Plane:

So far we were considering rectilinear motion of an object. The direction of motion of the object was always along one straight line. Now we will consider the motion of an object in two dimensions, i.e., along a plane. Here, the direction of the force acting on an object will not be in the same line as its initial velocity. Thus, the velocity and acceleration will have different directions. For this reason we have to use vector equations. The definitions of various terms given in section 3.2 will remain valid except that the magnitude of the average velocity and the value of average speed will be different as the magnitude of the displacement need not be equal to the path length. For example, if a particle travels along a circle and comes back to its original position, its displacement will be zero but the path length will be equal to the circumference of the circle.

3.3.1 Average and Instantaneous Velocities:

For studying the motion of an object in two dimensions, for simplicity, we will take the plane to be the *x*-*y* plane. To describe the position of an object in this plane we will have to specify, both its *x* and *y* coordinates. The definitions of displacement, average and instantaneous velocities, average and instantaneous speeds and acceleration will be the same as those for rectilinear motion except that each of these quantities will now have components along the *x* and *y* directions. Let us assume the object to be at point P at time t_1 as shown in Fig. 3.4 (a).

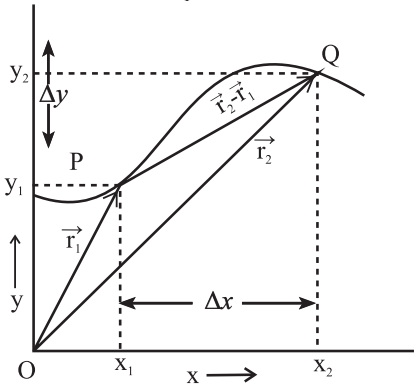


Fig. 3.4 (a) Motion in two dimensions

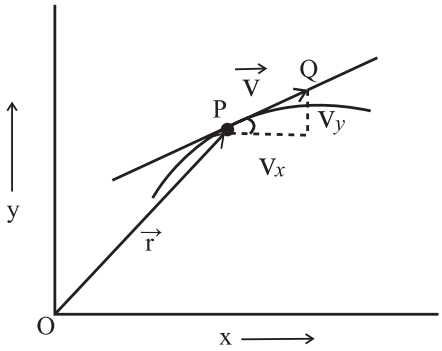


Fig. 3.4 (b) Instantaneous velocity

The position of the object will be described by its position vector $\vec{r}_1$. This can be written in terms of its components along the *x* and *y* axes as

$$\vec{r}_1 = x_1\hat{i} + y_1\hat{j} \quad \text{--- (3.12)}$$

At time t_2, let the position of the object be Q and its position vector be $\vec{r}_2$

$$\vec{r}_2 = x_2\hat{i} + y_2\hat{j} \quad \text{--- (3.13)}$$

The displacement of the particle from t_1 to t_2 shown by PQ, i.e., in time $t = t_2 - t_1$ is given by

$$\overrightarrow{\Delta r} = \vec{r}_2 - \vec{r}_1 = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} \quad \text{--- (3.14)}$$

We can write the average velocity of the object as

$$\vec{v}_{av} = \frac{\Delta\vec{r}}{\Delta t} = \left(\frac{x_2 - x_1}{t_2 - t_1}\right)\hat{i} + \left(\frac{y_2 - y_1}{t_{21} - t}\right)\hat{j}$$

$$\vec{v}_{av} = (v_{av})_x\hat{i} + (v_{av})_y\hat{j} \quad \text{--- (3.15)}$$

where, $(v_{av})_x = (x_2\text{-}x_1)/(t_2 - t_1)$ and

$$(v_{av})_y = (y_2\text{-}y_1)/(t_2 - t_1) \quad \text{--- (3.16)}$$

Average velocity is a vector whose direction is along $\Delta\vec{r}$ (see Eq. (3.2)), i.e., along the direction of displacement. In terms of its components, the magnitude (v) and direction (the angle θ that the velocity vector makes with the x-axis) can be written as (see Chapter 2)

$$v_{av} = \sqrt{(v_{av})_x^2 + (v_{av})_y^2} \quad \text{and}$$

$$\tan\theta = (v_{av})_y/(v_{av})_x \quad \text{--- (3.17)}$$

Figure 3.4(b) shows the trajectory of an object moving in two dimensions. The instantaneous velocity of the object at point P along the trajectory is along the tangent to the curve at P. This is shown by the vector PQ. Its x and y components v_x and v_y are also shown in the figure.

The instantaneous velocity of the object can be written in terms of derivative as (see Eq. 3.3)

$$\vec{v} = \lim_{\Delta t\to 0}\left(\frac{\Delta\vec{r}}{\Delta t}\right) = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}\right)\hat{i} + \left(\frac{dy}{dt}\right)\hat{j} \text{-- (3.18)}$$

The magnitude and direction of the instantaneous velocity are given by

$$v = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}, \quad \text{--- (3.19)}$$

$$\tan\theta = (dy/dt)/(dx/dt) = dy/dx \quad \text{--- (3.20)}$$

which is the slope of the tangent to the curve at the point at which we are calculating the instantaneous velocity.

3.3.2 Average and Instantaneous Acceleration:

Again, the definitions are the same as those for rectilinear motion. Thus, the average acceleration ($\vec{a}_{av}$) of a particle between times t_1 and t_2 can be written as

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \left(\frac{v_{2x} - v_{1x}}{t_2 - t_1}\right)\hat{i} + \left(\frac{v_{2y} - v_{1y}}{t_2 - t_1}\right)\hat{j} \text{ - (3.21)}$$

where $\vec{v}_2$ and $\vec{v}_1$ are the velocities of the particle at times t_2 and t_1 respectively.

$$\vec{a}_{av} = (a_{av})_x\hat{i} + (a_{av})_y\hat{j} \quad \text{--- (3.22),}$$

$(a_{av})_x$ and $(a_{av})_y$ being the x and y components of the average acceleration.

The magnitude and direction of the acceleration are given by

$$a_{av} = \sqrt{(a_{av})_x^2 + (a_{av})_y^2} \quad \text{--- (3.23)}$$

and

$$\tan\theta = (a_{av})_y/(a_{av})_x \quad \text{--- (3.24)}$$

The instantaneous acceleration is given by (see Eq. (3.5))

$$\vec{a} = \lim_{t\to 0}\left(\frac{\Delta\vec{v}}{\Delta t}\right) = \frac{d\vec{v}}{dt} = \left(\frac{dv_x}{dt}\right)\hat{i} + \left(\frac{dv_y}{dt}\right)\hat{j} \text{ -(3.25)}$$

$$= \frac{d}{dt}\left(\frac{dx}{dt}\right)\hat{i} + \frac{d}{dt}\left(\frac{dy}{dt}\right)\hat{j} = \left(\frac{d^2x}{dt^2}\right)\hat{i} + \left(\frac{d^2y}{dt^2}\right)\hat{j}$$

--- (3.26)

Thus, the x and y components of the instantaneous acceleration are respectively given by

$a_x = d^2x/dt^2$ and $a_y = d^2y/dt^2$ --- (3.27)

The magnitude and direction of the instantaneous acceleration are given by

$$a = \sqrt{\left(\frac{d^2x}{dt^2}\right)^2 + \left(\frac{d^2y}{dt^2}\right)^2} \quad \text{--- (3.28),}$$

and

$$\tan\theta = (dv_y/dt)/(dv_x/dt) = dv_y/dv_x \quad \text{--- (3.29)}$$

which is the slope of the tangent to the curve in velocity graph, i.e., a plot of v_y versus v_x.

Example 3.4: The position vectors of three particles are given by

$\vec{x}_1 = (5\hat{i} + 5\hat{j})$ m, $\vec{x}_2 = (5t\,\hat{i} + 5t\,\hat{j})$ m and $\vec{x}_3 = (5t\,\hat{i} + 10t^2\,\hat{j})$m as a function of time t. Determine the velocity and acceleration for

each, in SI units.

Solution: $\vec{v}_1 = d\vec{x}_1/dt = 0$ as $\vec{x}_1$ does not depend on time t.

Thus, the particle is at rest.

$\vec{v}_2 = d\vec{x}_2/dt = 5\hat{i} + 5\hat{j}$ m/s. $\vec{v}_2$ does not change with time. $\therefore \vec{a}_2 = 0$

$v_2 = \sqrt{5^2+5^2} = 5\sqrt{2}$ m/s, $\tan\theta = 5/5 = 1$ or $\theta = 45°$. Thus, the direction of v_2 makes an angle of 45° to the horizontal.

$\vec{v}_3 = d\vec{x}_3/dt = 5\hat{i} + 20t\hat{j}$.

$\therefore v_3 = \sqrt{5^2+(20t)^2}$ m/s. Its direction is along $\theta = \tan^{-1}\left(\frac{20t}{5}\right)$ with the horizontal.

$\vec{a}_3 = \frac{d\vec{v}_3}{dt} = 20\hat{j}$ m/s^2

Thus, the particle 3 is getting accelerated along the *y*-axis at 20 m/s^2.

3.3.3 Equations of Motion for an Object travellinging a Plane with Uniform Acceleration:

We have derived equations of motion for an object in rectilinear motion in section 3.2. We will now derive similar equations for a particle moving with uniform acceleration in two dimensions. Let the initial velocity of the object be $\vec{u}$ at $t = 0$ and its velocity at time t be $\vec{v}$. As the acceleration is constant, the average acceleration and the instantaneous acceleration will be equal. By using the definition of acceleration (Eq. (3.21)), we get

$$\vec{a} = (\vec{v} - \vec{u})/(t - 0)$$

$$\text{or } \vec{v} = \vec{u} + \vec{a}t \qquad \text{--- (3.30)}$$

which is the same as Eq. (3.7) but is in vector form.

Let the displacement from time $t = 0$ to t be $\vec{s}$. This can be calculated from the average velocity of the object during this time. For constant acceleration, $\vec{v}_{av} = \frac{\vec{u}+\vec{v}}{2}$

$$\therefore \vec{s} = (\vec{v}_{av})t = \left(\frac{\vec{u}+\vec{v}}{2}\right)t = \left(\frac{\vec{u}+\vec{u}+\vec{a}t}{2}\right)t$$

$$\therefore \vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2 \qquad \text{--- (3.31),}$$

which is the vector form of Eq. (3.8).

Eq. (3.30) and (3.31) can be resolved into their x and y components so as to get corresponding scalar equations as follows.

$$v_x = u_x + a_x t \qquad \text{--- (3.32)}$$

and $$v_y = u_y + a_y t \qquad \text{--- (3.33)}$$

$$s_x = u_x t + \frac{1}{2}a_x t^2 \qquad \text{--- (3.34)}$$

and $$s_y = u_y t + \frac{1}{2}a_y t^2 \qquad \text{--- (3.35)}$$

We can see that Eqs. (3.32) and (3.34) involve only the x components of displacement, velocity and acceleration while Eqs. (3.33) and (3.35) involve only the y components of these quantities. Thus the two sets of equations are independent of each other and can be solved independently. We can thus see that the motion along the x direction of an object is completely controlled by the x components of velocity and acceleration while that along the y direction is completely controlled by the y components of these quantities. This makes it easy to study the motion in two dimensions which gets converted to two independent rectilinear motions along two perpendicular directions.

Always Remember:

Motion in two dimensions can be resolved into two independent motions in mutually perpendicular directions.

Example 3.5: The initial velocity of an object is $\vec{u} = 5\hat{i} + 10\hat{j}$ m/s. Its constant acceleration is $\vec{a} = 2\hat{i} + 3\hat{j}$ m/s^2. Determine the velocity and the displacement after 5 s.

Solution:

$$\vec{v} = \vec{u} + \vec{a}t$$

$$= (5\hat{i} + 10\hat{j}) + (2\hat{i} + 3\hat{j})(5) = 15\hat{i} + 25\hat{j}$$

$$\therefore v = \sqrt{v_x^2 + v_y^2}$$

$$= \sqrt{15^2 + 25^2} = \sqrt{225 + 625} = \sqrt{850}$$

$$= 29.15 \text{ m/s}$$

Direction of $\vec{v}$ with *x*-axis is $\tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}$

$$\left(\frac{25}{15}\right) = \tan^{-1}(1.667) = 59^{o}$$

$$\vec{s} = \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2$$

$$= \left(5\hat{i} + 10\hat{j}\right)(5) + \frac{1}{2}\left(2\hat{i} + 3\hat{j}\right)5^2$$

$$= 50\hat{i} + (87.5)\,\hat{j}$$

$$\therefore s = \sqrt{s_x^2 + s_y^2} = \sqrt{50^2 + 87.5^2}$$

$$= \sqrt{2500 + 7656.25}$$

$$= \sqrt{10156.25} = 100.78 \text{ m}$$

at $\tan^{-1}\frac{87.5}{50} = 60°15'$ with x-axis.

3.3.4 Relative Velocity:

Relative velocity between two objects moving in a plane can be defined in a way similar to that for objects moving along a straight line. The relative velocity of object A having velocity $\vec{v}_A$, with respect to the object B having velocity $\vec{v}_B$, is given by

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B \qquad \text{--- (3.36)}$$

Similarly, the relative velocity of object B with respect to object A, is given by

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A \qquad \text{--- (3.37)}$$

We can see that the magnitudes of the two relative velocities (v_{AB} and v_{BA}) are equal and their directions are opposite.

Consider a number of objects A, B, C, D ---- Y, Z, moving with respect to the other. Using the symbol v_{AB} for representing the velocity of A relative to B etc, the velocity of A relative to Z can be written as

$$\vec{v}_{AZ} = \vec{v}_{AB} + \vec{v}_{BC} + \vec{v}_{CD} + \ldots + \vec{v}_{XY} + \vec{v}_{YZ}$$

Note the order of subscripts (A→B→C→D---→Z).

Example 3.6: An aeroplane is travelling northward with a speed of 300 km/hr with respect to the Earth, when the wind is blowing from east to west at a speed of 100 km/hr. What is the velocity of the aeroplane with respect to the wind?

Solution: Let the velocity of the aeroplane with respect to Earth be $\vec{v}_{AE}$, velocity of wind with respect to Earth be $\vec{v}_{WE}$. The velocity of aeroplane with respect to wind, $\vec{v}_{AW}$ can be determined by the following expression:

$\vec{v}_{AW} = \vec{v}_{AE} + \vec{v}_{EW} = \vec{v}_{AE} - \vec{v}_{WE} = 100\,\hat{i} + 300\,\hat{j}$, considering north along +y axis.

Magnitude of $\vec{v}_{AW} = \sqrt{(10000 + 90000)}$

$= 100\sqrt{10}$ km/hr, and its direction,

$\theta = \tan^{-1}\left(\frac{300}{100}\right) = 71.6°$ is towards north of east.

3.3.5 Projectile Motion:

Any object in flight after being thrown with some velocity is called a projectile and its motion is called projectile motion. We often see projectile motion in our day-to-day life. Children throw stones towards trees for getting tamarind pods or mangoes. A bowler bowls a ball towards a batsman in cricket, a basket ball player throws a ball towards the basket, all these are illustrations of projectile motion. In this motion, we have objects (projectiles) with given initial velocity, moving under the influence of the Earth's gravitational field. The projectile has two components of velocity, one in the horizontal, i.e., along x-direction and the other in the vertical, i.e., along the y direction. The acceleration due to gravity acts only along the vertically downward direction. The horizontal component of velocity, therefore, remains unchanged as no force is acting in the horizontal direction, while the vertical component changes in accordance with laws of motion with $\vec{a}_x$ being 0 and $\vec{a}_y$ $(= -\vec{g})$ being the downward acceleration due to gravity (upward is positive). Unless stated otherwise, retarding forces like air resistance, etc., are neglected for the projectile motion.

Let us assume that the initial velocity of the projectile is $\vec{u}$ and its direction makes an angle θ with the horizontal as shown in Fig. 3.5. The projectile is thrown from the ground. We take the x-axis along the ground and y-axis in the vertical direction. The horizontal and vertical components of initial velocity are u

cosθ and u sinθ respectively. The horizontal component remains unchanged in absence of any force acting in that direction, while the vertical component changes according to (Eq. 3.33) with $a_y = -g$ and $u_y = u \sin\theta$.

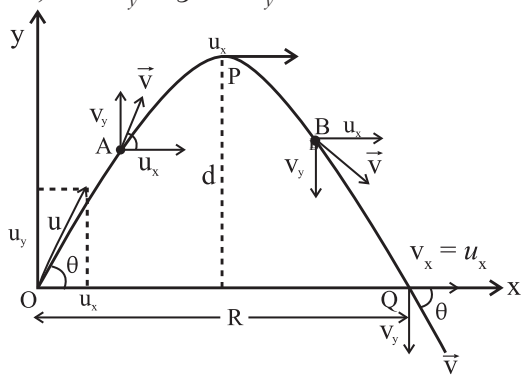


Fig.3.5: Trajectory of a projectile.

Thus, the components of velocity at time t are given by

$$v_x = u_x = u\cos\theta \qquad \text{--- (3.38)}$$

$$v_y = u_y - gt = u\sin\theta - gt \qquad \text{--- (3.39)}$$

As $0 < \theta < 90°$, the vertical component initially is in the upward direction. Similarly, the displacements of the projectile in the horizontal and vertical directions at time t, according to Eqs. (3.34) and (3.35) are given by

$$s_x = u\cos\theta.t \qquad \text{---- (3.40)}$$

$$s_y = u\sin\theta.t - \frac{1}{2}gt^2 \qquad \text{--- (3.41)}$$

The direction of motion of the projectile at any time t makes an angle α with the horizontal which is given by

$$\tan\alpha = v_y(t)/v_x(t) \qquad \text{--- (3.42)}$$

The vertical velocity keeps on decreasing as the projectile goes up and becomes zero at certain time. At that time the height of the projectile is maximum. The velocity then starts increasing in the downward direction as the particle is now falling under the Earth 's gravitational field with a constant horizontal component of velocity. After a while the projectile reaches the ground. The trajectory of the object is shown in Fig. 3.5. The projectile is assumed to start from the origin of the coordinate system, O. The point of maximum height is indicated by P and the point where it falls down to the ground is indicated by Q. The horizontal and vertical components of velocity are shown at these points as well as at two intermediate points A and B, on the trajectory of the projectile. Note that the horizontal component of velocity remains the same, i.e., u_x, while the vertical component decreases and becomes zero at P. After that it changes its direction, its magnitude increases and becomes equal to u_y again at Q. The horizontal distance covered by the projectile before it falls to the ground is OQ. We can derive the equation of the trajectory of the projectile as follows.

Let the time taken by the projectile to reach the maximum height be t_0. The trajectory of the object being symmetrical, it can be shown by using equations of motion, that the object will take the same time in going up in air and coming down to the ground. At the highest point P, $t = t_0$ and $v_y = 0$. Using Eq. (3.39),

we get, $0 = u\sin\theta - gt_0$

$$t_0 = (u\sin\theta)/g \qquad \text{--- (3.43)}$$

$\therefore$ Total time in air = $T = 2t_0$ is the ***time of flight***.

The total horizontal distance travelled by the particle in this time T can be obtained by using Eq. (3.40) as

$$R = u_x.\,T = u\cos\theta.2t_0 = u\cos\theta.\,(2u\sin\theta)/g$$

$$= 2\,u_x\,u_y/g = u^2(2\sin\theta\cos\theta)/g$$

$$= u^2\sin2\theta/g \qquad \text{--- (3.44)}$$

This maximum horizontal distance travelled by the projectile is called the ***horizontal range*** R of the projectile and depends on the magnitude and direction of initial velocity of the projectile as well as the value of acceleration due to gravity at that place.

For maximum horizontal range,

$\sin2\theta = 1 \;\therefore 2\theta = 90^0 \; or \; \theta = 45^0$

Hence, $R = R_{max} = \dfrac{u^2}{g}$ for $\theta = 45^0$

The ***maximum height*** H reached by the projectile, having certain value of θ, is the distance travelled along the vertical (y) direction in time t_0. This can be calculated by using Eq. (3.41) as

$$H = u\sin\theta.t_0 - \frac{1}{2}gt_0^2$$

$$= u\sin\theta\left(\frac{u\sin\theta}{g}\right) - \frac{1}{2}g\left(\frac{u\sin\theta}{g}\right)^2$$

$$= \frac{u^2\sin^2\theta}{2g} = \frac{u_y^2}{2g} \qquad \text{--- (3.45)}$$

Do you know ?

All the above expressions of T, R, R_{max} and H are valid if the entire motion is governed only by gravitational acceleration g, i.e., retarding forces like air resistance are absent. However, in reality, it is never so. As a result, time of ascent t_a and time of decent t_d are not equal but $t_a > t_d$. Also, in order to achieve maximum horizontal range for given initial velocity, the angle of projection should be greater than 45^0 and the range is much less than $\frac{u^2}{g}$.

Example 3.7: A stone is thrown with an initial velocity components of 20 m/s along the vertical, and 15 m/s along the horizontal direction. Determine the position and velocity of the stone after 3 s. Determine the maximum height that it will reach and the total distance travelled along the horizontal on reaching the ground. (Assume $g = 10$ m/s^2)

Solution: The initial velocity of the stone in x-direction $= u\cos\theta = 15$ m/s and in y-direction $= u\sin\theta = 20$ m/s.

After 3 s, $v_x = u\cos\theta = 15$ m/s and $v_y = u\sin\theta - gt = 20 - 10(3) = -10$ m/s $= 10$ m/s downwards.

$$\therefore v = \sqrt{v_x^2 + v_y^2} = \sqrt{15^2 + 10^2}$$
$$= \sqrt{225 + 100} = \sqrt{325}$$
$$= 18.03 \text{ m/s}$$

$\tan\alpha = v_y / v_x = 10/15 = 2/3$

$\therefore \alpha = \tan^{-1}(2/3) = 33° 41'$ with the horizontal.

$s_x = (u\cos\theta)\, t = 15 \times 3 = 45$ m,

$s_y = (u\sin\theta)\, t - \frac{1}{2} gt^2 = 20 \times 3 - 5(3)^2 = 15$ m.

Thus the stone will be at a distance 45 m along horizontal and 15 m along vertical direction from the initial position after time 3 s. The velocity is 18.03 m/s making an angle 33° 41' with the horizontal.

The maximum vertical distance travelled is given by $H = (u\sin\theta)^2/(2g) = 20^2/(2 \times 10) = 20$ m

Maximum horizontal distance travelled

$R = 2.u_x.u_y/g = 2(15)(20)/10 = 60$ m

Equation of motion for a projectile

We can derive the equation of motion of the projectile which is the relation between the displacements of the projectile along the vertical and horizontal directions. This can be obtained by eliminating t between the equations giving these displacements, i.e., Eqs. (3.40) and (3.41).

As the projectile starts from $\vec{x} = 0$, we can write $s_x = x$ and $s_y = y$.

$$\therefore s_x = (u\cos\theta)t \quad \therefore t = \frac{s_x}{u\cos\theta} = \frac{x}{u\cos\theta}$$

$$\therefore y = (u\sin\theta)t - \frac{1}{2} gt^2$$

$$= (u\sin\theta)\left(\frac{x}{u\cos\theta}\right) - \frac{1}{2} g\left(\frac{x}{u\cos\theta}\right)^2$$

$$\therefore y = (\tan\theta)x - \frac{1}{2}\left(\frac{g}{u^2\cos^2\theta}\right)x^2 \qquad \text{--- (3.46)}$$

This is the equation of the trajectory of the projectile. Here, u and θ are constants for the given projectile motion. The above equation is of the form

$$y = Ax + Bx^2 \qquad \text{--- (3.47)}$$

which is the equation of a parabola. Thus, the path, i.e., the trajectory of a projectile is a parabola.

3.4 Uniform Circular Motion:

An object moving with *constant speed* along a circular path is said to be in *uniform* circular motion (UCM). Such a motion is only possible if its velocity is *always tangential* to its circular path, *without change in its magnitude*.

To change the direction of velocity, acceleration is a must. However, if the acceleration or its component is in line with the velocity (along or opposite to the velocity), it will *always* change the speed (magnitude of velocity) in which case it will not continue its uniform circular motion. In order to achieve both these requirements, the acceleration must be (i) perpendicular to the tangential velocity, (ii) of constant magnitude and (iii) always directed

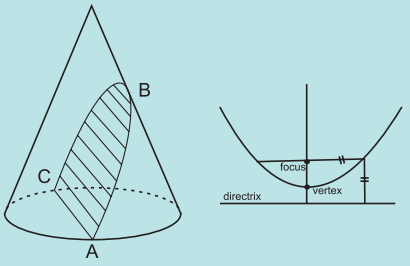
towards the centre of the circular trajectory. Such an acceleration is called *centripetal* (centre seeking) acceleration and the force causing this acceleration is *centripetal* force.

Thus, in order to realize a circular motion, there are two requirements; (i) *tangential velocity* and (ii) *centripetal force* of *suitable constant magnitude*.

An example is the motion of the moon going around the Earth in an early circular orbit as a result of the constant gravitational attraction of fixed magnitude felt by it towards the Earth.

Do you know ?

A parabola is a symmetrical open curve obtained by the intersection of a cone with a plane which is parallel to its side. Mathematically, the parabola is described with the help of a point called the focus and a straight line called the directrix shown in the accompanying figure. The parabola is the locus of all points which are equidistant from the focus and the directrix. The chord of the parabola which is parallel to the directrix and passes through the focus is called latus rectum of the parabola as shown in the accompanying figure.



3.4.1 Period, Radius Vector and Angular Speed:

Consider an object of mass m, moving with a uniform speed v, along a circle of radius r. Let T be the time period of revolution of the object, i.e., the time taken by the object to complete one revolution or to travel a distance of $2\pi r$.

Thus, $T = 2\pi r/\text{v}$

$$\therefore \text{Speed v} = \frac{\text{Distance}}{\text{Time}} = \frac{2\pi r}{T} \quad \text{--- (3.48).}$$

During circular motion of a point object, the position vector of the object from centre of the circle is the radius vector $\vec{r}$. Its magnitude is radius r and it is directed away from the centre to the particle, i.e., away from the centre of the circle. As the particle performs UCM, this radius vector describes equal angles in equal intervals of time. At this stage we can define a new quantity called angular speed ω which gives the angle described by the radius vector, per unit time. It is analogous to speed which is distance travelled per unit time.

During one complete revolution, the angle described is 2π and the time taken is period T. Hence, the angular speed

$$\omega = \frac{\text{Angle}}{\text{time}} = \frac{2\pi}{T} = \frac{(2\pi)}{\left(\frac{2\pi r}{\text{v}}\right)} = \frac{\text{v}}{r} \quad \text{--- (3.49)}$$

The unit of ω is radian/sec.

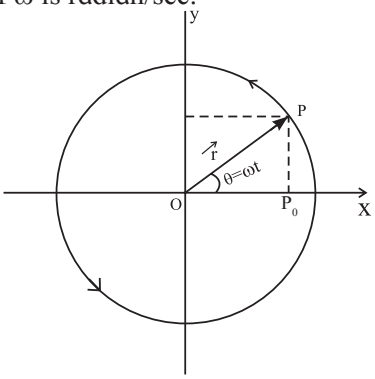


Fig.3.6: Uniform circular motion.

3.4.2 Expression for Centripetal Acceleration:

Figure 3.6 shows a particle P performing a UCM in anticlockwise sense along a circle of radius r with angular speed ω and period T. Let us choose the coordinates such that this motion is in the xy- plane having centre at the origin O. Initially (for simplicity), let the particle be at P_0 on the positive x-axis. At a given instant t, the radius vector of P makes an angle θ with the x-axis.

$$\therefore \theta = \omega t \text{ and so } \frac{d\theta}{dt} = \omega$$

x and y components of the radius vector $\vec{r}$ will then be $r\cos\theta$ and $r\sin\theta$ respectively.

$$\therefore \vec{r} = (r\cos\theta)\hat{i} + (r\sin\theta)\hat{j}$$

$$= (r\cos[\omega t])\hat{i} + (r\sin[\omega t])\hat{j} \quad \text{--- (3.50)}$$

Time derivative of position vector $\vec{r}$ gives

instantaneous velocity $\vec{v}$ and time derivative of velocity $\vec{v}$ gives instantaneous acceleration $\vec{a}$. Magnitudes of r and ω are constants.

$$\therefore \vec{v} = \frac{d\vec{r}}{dt} = r\left(-\omega\sin[\omega t]\hat{i} + \omega\cos[\omega t]\hat{j}\right)$$
$$= r\omega\left(-\sin[\omega t]\hat{i} + \cos[\omega t]\hat{j}\right)$$

--- (3.51)

$$\therefore \vec{a} = \frac{d\vec{v}}{dt} = r\omega\left(-\omega\cos[\omega t]\hat{i} - \omega\sin[\omega t]\hat{j}\right)$$
$$= -\omega^2\left(r\cos[\omega t]\hat{i} + r\sin[\omega t]\hat{j}\right) = -\omega^2\vec{r}$$

--- (3.52)

Here minus sign shows that the acceleration is opposite to that of $\vec{r}$, i.e., towards the centre. This is the centripetal acceleration.

The magnitude of acceleration,

$$a = \omega^2 r = \frac{v^2}{r} = \omega v \qquad \text{--- (3.53)}$$

The force providing this acceleration should also be along the same direction, hence centripetal.

$$\therefore \vec{F} = m\vec{a} = -m\omega^2\vec{r} \quad \text{--- (3.54)}$$

Magnitude of $F = m\omega^2 r = \frac{mv^2}{r} = m\omega v$ - (3.55)

Conical pendulum

In a simple pendulum a mass m is suspended by a string of length l and moves along an arc of a vertical circle. If the mass instead revolves in a horizontal circle and the string which makes a constant angle with the vertical describes a cone whose vertex is the fixed point O, then mass-string system is called a conical pendulum as shown in Fig. 3.7. In the absence of friction, the system will continue indefinitely once started.

As shown in the figure, the forces acting on the bob of mass, m, of the conical pendulum are: (i) Gravitational force, mg, acting vertically downwards, (ii) Force due to tension $\vec{T}$ acting along the string directed towards the support. These are the only two forces acting on the bob.

For the bob to undergo horizontal circular motion, (on a circle of radius r) the resultant force must be centripetal, (directed towards the centre of the circle). In other words vertical gravitational force must be balanced.

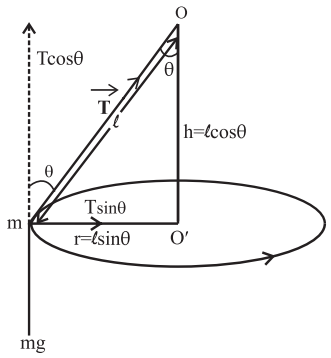


Fig 3.7: Conical pendulum

Thus, we resolve tension $\vec{T}$ into two mutually perpendicular components. Let θ be the angle made by the string with the vertical at any position. The component T cos θ is acting vertically upwards. The inclination should be such that T cos θ = mg, so that there is no net vertical force.

The resultant force on the bob is then T sin θ which is radial or centripetal or directed towards centre O'.

$$\therefore \quad T\sin\theta = mv^2/r$$

$$\tan\theta = \frac{T\sin\theta}{T\cos\theta} = \frac{(mv^2/r)}{mg} = \frac{v^2}{rg}$$

Since we know $v = \frac{2\pi r}{T}$

$$\therefore \tan\theta = \frac{4\pi^2 r^2}{T^2 rg}$$

$$T = 2\pi\sqrt{\frac{r}{g\tan\theta}}$$

$$T = 2\pi\sqrt{\frac{l\sin\theta}{g\tan\theta}} \quad (\because r = l\sin\theta)$$

$$T = 2\pi\sqrt{\frac{l\cos\theta}{g}}$$

$$T = 2\pi\sqrt{\frac{h}{g}} \quad (\because h = l\cos\theta) \qquad \text{--- (3.56)}$$

where l is length of the pendulum and h is the vertical distance of the horizontal circle from the fixed point O.

Example 3.8: An object of mass 50 g moves uniformly along a circular orbit with an angular speed of 5 rad/s. If the linear speed of the particle is 25 m/s, what is the radius of the

circle? Calculate the centripetal force acting on the particle.

Do you know ?

1. The centripetal force is not one of the external forces acting on the object. As can be seen from above, the actual forces acting on the bob are T and mg, the resultant of these is the centripetal force. Conversely, if the resultant force is centripetal, motion must be circular.
2. In planetary motion, the gravitational force between Sun and the planets provides the necessary centripetal force for the circular motion.

Solution: The linear speed and angular speed are related by $v = \omega r$

$\therefore r = v/\omega = 25/5$ m = 5 m.

Centripetal force acting on the object = $\frac{mv^2}{r} = \frac{0.05 \times 25^2}{5} = 6.25$ N.

Example 3.9: An object is travelling in a horizontal circle with uniform speed. At $t = 0$, the velocity is given by $\vec{u} = 20\hat{i} + 35\hat{j}$ km/s. After one minute the velocity becomes $\vec{v} = -20\hat{i} - 35\hat{j}$. What is the magnitude of the acceleration?

Solution: Magnitude of initial and final velocities =

$= u = \sqrt{(20)^2 + (35)^2}$ m / s

$= \sqrt{1625}$ m / s

$= 40.3$ m / s

As the velocity reverses in 1 min, the time period of revolution is 2 min.

$T = \frac{2\pi r}{u}$, giving $r = \frac{uT}{2\pi}$

$a = \frac{u^2}{r} = \frac{u^2 2\pi}{uT} = \frac{2\pi u}{T} = \frac{2 \times 3.14 \times 40.3}{2 \times 60}$

$= 2.11$ m s^{-2}

Internet my friend

1. hyperphysics.phy-astr.gsu.edu/hbase/mot.html#motcon
2. www.college-physics.com/book/mechanics

Exercises

1. Choose the correct option.

i) An object thrown from a moving bus is on example of
(A) Uniform circular motion
(B) Rectilinear motion
(C) Projectile motion
(D) Motion in one dimension

ii) For a particle having a uniform circular motion, which of the following is constant
(A) Speed (B) Acceleration
(C) Velocity (D) Displacement

iii) The bob of a conical pendulum under goes
(A) Rectilinear motion in horizontal plane
(B) Uniform motion in a horizontal circle
(C) Uniform motion in a vertical circle
(D) Rectilinear motion in vertical circle

iv) For uniform acceleration in rectilinear motion which of the following is not correct?
(A) Velocity-time graph is linear
(B) Acceleration is the slope of velocity time graph
(C) The area under the velocity-time graph equals displacement
(D) Velocity-time graph is nonlinear

v) If three particles A, B and C are having velocities $\vec{v}_A$, $\vec{v}_B$ and $\vec{v}_C$ which of the following formula gives the relative velocity of A with respect to B
(A) $\vec{v}_A + \vec{v}_B$ (B) $\vec{v}_A - \vec{v}_C + \vec{v}_B$

(C) $\vec{v}_A - \vec{v}_B$ (D) $\vec{v}_C - \vec{v}_A$

2. Answer the following questions.

i) Separate the following in groups of scalar and vectors: velocity, speed, displacement, work done, force, power, energy, acceleration, electric charge, angular velocity.

ii) Define average velocity and instantaneous velocity. When are they same?

iii) Define free fall.

iv) If the motion of an object is described by $x = f(t)$ write formulae for instantaneous velocity and acceleration.

v) Derive equations of motion for a particle moving in a plane and show that the motion can be resolved in two independent motions in mutually perpendicular directions.

vi) Derive equations of motion graphically for a particle having uniform acceleration, moving along a straight line.

vii) Derive the formula for the range and maximum height achieved by a projectile thrown from the origin with initial velocity $\vec{u}$ at an angel θ to the horizontal.

viii) Show that the path of a projectile is a parabola.

ix) What is a conical pendulum? Show that its time period is given by $2\pi\sqrt{\frac{l\cos\theta}{g}}$, where l is the length of the string, θ is the angle that the string makes with the vertical and g is the acceleration due to gravity.

x) Define angular velocity. Show that the centripetal force on a particle undergoing uniform circular motion is $-m\omega^2\vec{r}$.

3. Solve the following problems.

i) An aeroplane has a run of 500 m to take off from the runway. It starts from rest and moves with constant acceleration to cover the runway in 30 sec. What is the velocity of the aeroplane at the take off ?

[Ans: 120 km/hr]

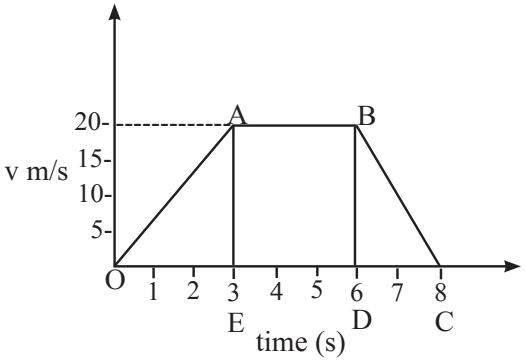
ii). A car moving along a straight road with a speed of 120 km/hr, is brought to rest by applying brakes. The car covers a distance of 100 m before it stops. Calculate (i) the average retardation of the car (ii) time taken by the car to come to rest.

[Ans: 50/9 m/sec^2, 6 sec]

iii) A car travels at a speed of 50 km/hr for 30 minutes, at 30 km/hr for next 15 minutes and then 70 km/hr for next 45 minutes. What is the average speed of the car?

[Ans: 56.66 km/hr]

iv) A velocity-time graph is shown in the adjoining figure.



Determine:

(i) initial speed of the car (ii) maximum speed attained by the car (iii) part of the graph showing zero acceleration (iv) part of the graph showing constant retardation (v) distance travelled by the car in first 6 sec.

[Ans: (i) 0 (ii) 20 m/sec (iii) AB (iv) BC (v) 90 m]

v) A man throws a ball to maximum horizontal distance of 80 meters. Calculate the maximum height reached.

[Ans: 20 m]

vi) A particle is projected with speed v_0 at angle θ to the horizontal on an inclined surface making an angle ϕ ($\phi < \theta$) to the horizontal. Find the range of the projectile along the inclined surface.

[Ans: $R = \frac{2v_0^2\cos\theta\sin(\theta-\phi)}{g\cos^2\phi}$]

vii) A metro train runs from station A to B to C. It takes 4 minutes in travelling from station A to station B. The train halts at station B for 20 s. Then it starts from station B and reaches station C in next 3 minutes. At the start, the train accelerates for 10 sec to reach the constant speed of 72 km/hr. The train moving at the constant speed is brought to rest in 10 sec. at next station. (i) Plot the velocity- time graph for the train travelling from the station A to B to C. (ii) Calculate the distance between the stations A, B and C.

[Ans: AB = 4.6 km, BC =3.4 km]

viii) A train is moving eastward at 10 m/sec. A waiter is walking eastward at 1.2m/sec; and a fly is flying toward the north across the waiter's tray at 2 m/s. What is the velocity of the fly relative to Earth

[Ans: 11.4 m/s, 10° due north of east]

ix) A car moves in a circle at the constant speed of 50 m/s and completes one revolution in 40 s. Determine the magnitude of acceleration of the car.

[Ans: 7.85 m s^{-2}]

x) A particle moves in a circle with constant speed of 15 m/s. The radius of the circle is 2 m. Determine the centripetal acceleration of the particle.

[Ans: 112.5 m s^{-2}]

xi) A projectile is thrown at an angle of 30° to the horizontal. What should be the range of initial velocity (u) so that its range will be between 40m and 50 m? Assume g = 10 m s^{-2}.

[Ans: $21.49 \leq u \leq 24.03$ m s^{-2}]

***

4. Laws of Motion



Can you recall?

1. What are different types of motions?
2. What do you mean by kinematical equations and what are they?
3. Newton's laws of motion apply to most bodies we come across in our daily lives.
4. All bodies are governed by Newton's law of gravitation. Gravitation of the Earth results into weight of objects.
5. Acceleration is directly proportional to force for fixed mass of an object.
6. Bodies possess potential energy and kinetic energy due to their position and motion respectively which may change. Their total energy is conserved in absence of any external force.

4.1. Introduction:

If an object continuously changes its position, it is said to be in motion. Mechanics is a branch of Physics that deals with motion. There are basically two branches of mechanics (i) Statics, where we deal with objects at rest or in equilibrium under the action of balanced forces and (ii) Kinetics, which deals with actual motion.

Kinetics can be further divided into two branches **(i) Kinematics:** In kinematics, we describe various motions without discussing their cause. Various parameters discussed in kinematics are distance, displacement, speed, velocity and acceleration. **(ii) Dynamics:** In dynamics we describe the motion along with its cause, which is force and/or torque. Parameters discussed in dynamics are momentum, force, energy, power, etc. in addition to those in kinematics.

It must be understood that motion is strictly a relative concept, i.e., it should always be described in context to a reference frame. For example, if you are in a running bus, neither you nor your co-passengers sitting in the bus are in motion in your reference, i.e., moving bus. However, from the ground reference, bus, you and all the passengers are in motion.

If not random, motions in real life may be understood separately as linear, circular or rotational, oscillatory, etc., or some combinations of these. While describing any of these, we need to know the corresponding forces responsible for these motions. Trajectory of any motion is decided by acceleration $\vec{a}$ and the initial velocity $\vec{u}$.

a) Linear motion: Initial velocity may be zero or non-zero. If initial velocity is zero (starting from rest), acceleration in any direction will result into a linear motion.

   If initial velocity is not zero, the acceleration must be in line with the initial velocity (along the same or opposite direction to that of the initial velocity) for resultant motion to be linear.

b) Circular motion: If initial velocity is not zero and acceleration is throughout perpendicular to the velocity, the resultant motion will be circular.

c) Parabolic motion: If acceleration is constant and initial velocity is *not* in line with the acceleration, the motion is parabolic, e.g., trajectory of a projectile motion.

d) Other combinations of $\vec{u}$ and $\vec{a}$ will result into different more complicated motion.

4.2. Aristotle's Fallacy:

Aristotle (384BC-322BC) stated that "an external force is required to keep a body in uniform motion". This was probably based on a common experience like a ball rolling on a surface stops after rolling through some distance. Thus, to keep the ball moving with constant velocity, we have to continuously apply a force on it. Similar examples can be found elsewhere, like a paper plane flying through air or a paper boat propelled with some initial velocity.

Correct explanation to Aristotle's fallacy was first given by Galileo (1564-1642), which was later used by Newton (1643-1727) in

formulating *laws of motion*. Galileo showed that all the objects stop moving because of some resistive or opposing forces like friction, viscous drag, etc. In these examples such forces are frictional force for rolling ball, viscous drag or viscous force of air for paper plane and viscous force of water for the boat.

Thus, in reality, for an uninterrupted motion of a body an additional external force is required for overcoming these opposing forces.



Can you tell?

1. Was Aristotle correct?
2. If correct, explain his statement with an illustration.
3. If wrong, give the correct modified version of his statement.

4.3. Newton's Laws of Motion:

First law: Every inanimate object continues to be in its state of rest or of uniform *unaccelerated* motion unless and until it is acted upon by an *external, unbalanced* force.

Second law: Rate of change of linear momentum of a rigid body is directly proportional to the applied force and takes place in the direction of the applied force. On selecting suitable units, it takes the form $\vec{F} = \frac{d\vec{p}}{dt}$ (where $\vec{F}$ is the force and $\vec{p} = m\vec{v}$ is the linear momentum.

Third law: To every action (force), there is an equal and opposite reaction (force).

Discussion: From Newton's second law of motion, $\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v})$. For a *given* body, mass m is constant.

$$\therefore \vec{F} = m\frac{d\vec{v}}{dt} = m\vec{a} \ \ldots \text{(for constant mass)}$$

Thus, if $\vec{F} = 0$, $\vec{v}$ is constant. Hence if there is no force, velocity will not change. This is nothing but Newton's first law of motion.



Can you tell?

What is then special about Newton's first law if it is derivable from Newton's second law?

4.3.1. Importance of Newton's First Law of Motion:

(i) It shows an equivalence between '*state of rest*' and '*state of uniform motion along a straight line*' as both need a net unbalanced force to change the state. Both these are referred to as '*state of motion*'. The distinction between state of rest and uniform motion lies in the choice of the '*frame of reference*'.

(ii) It defines *force* as an entity (or a physical quantity) that brings about a change in the '*state of motion*' of a body, i.e., force is something that initiates a motion or controls a motion. Second law gives its quantitative understanding or its mathematical expression.

(iii) It defines *inertia* as a fundamental property of every physical object by which the object *resists* any change in its state of motion. Inertia is measured as the mass of the object. More specifically it is called inertial mass, which is the ratio of net force ($|\vec{F}|$) to the corresponding acceleration ($|\vec{a}|$).

4.3.2. Importance of Newton's Second Law of Motion:

(i) It gives mathematical formulation for quantitative measure of force as rate of change of linear momentum.



Do you know ?

Mathematical expression for force must be remembered as $\vec{F} = \frac{d\vec{p}}{dt}$ and ***not*** as $\vec{F} = m\vec{a}$

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{dm}{dt}(\vec{v}) + m\left(\frac{d\vec{v}}{dt}\right)$$

$$= \frac{dm}{dt}(\vec{v}) + m\vec{a}$$

For a *given* body, mass is constant, i.e., $\frac{dm}{dt} = 0$ and only in this case, $\vec{F} = m\vec{a}$

In the case of a rocket, both the terms are needed as both mass and velocity are varying.

(ii) It defines *momentum* $\left(\vec{p} = m\vec{v}\right)$ instead of velocity as the fundamental quantity related to motion. What is changed by a force is the momentum and not necessarily the velocity.

(iii) Aristotle's fallacy is overcome by considering *resultant unbalanced* force.

4.3.3 Importance of Newton's Third Law of Motion:

(i) It defines *action* and *reaction* as a pair of equal and opposite forces acting along the same line.

(ii) Action and reaction forces are always on *different* objects.

Consequences:

Action force exerted by a body x on body y, conventionally written as $\vec{F}_{yx}$, is the force experienced by y.

As a result, body y exerts *reaction force* $\vec{F}_{xy}$ on body x.

In this case, body x experiences the force $\vec{F}_{xy}$ only while the body y experiences the force $\vec{F}_{yx}$ only.

Forces $\vec{F}_{xy}$ and $\vec{F}_{yx}$ are equal in magnitude and opposite in their directions, but there is no question of cancellation of these forces as those are experienced by *different* objects.

Forces $\vec{F}_{xy}$ and $\vec{F}_{yx}$ *need not* be contact forces. Repulsive forces between two magnets is a pair of action-reaction forces. In this case the two magnets are not in contact. Gravitational force between Earth and moon or between Earth and Sun are also similar pairs of non-contact action-reaction forces.

Example 4.1: A hose pipe used for gardening is ejecting water horizontally at the rate of 0.5 m/s. Area of the bore of the pipe is 10 cm^2. Calculate the force to be applied by the gardener to hold the pipe *horizontally* stationary.

Solution: If ejecting water horizontally is considered as action force on the water, the water exerts a backward force (called recoil force) on the pipe as the reaction force.

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d\left(m\vec{v}\right)}{dt} = \frac{dm}{dt}\vec{v} + m\frac{d\vec{v}}{dt}$$

As v, the velocity of ejected water is constant, $F = \frac{dm}{dt}\vec{v}$, where $\frac{dm}{dt}$ is the rate at which mass of water is ejected by the pipe.

As the force is in the direction of velocity (horizontal), we can use scalars. $\therefore F = \frac{dm}{dt}v$

$$\frac{dm}{dt} = \frac{d\left(V\rho\right)}{dt} = \frac{d\left(Al\rho\right)}{dt} = A\rho\frac{dl}{dt} = A\rho v$$

where V = volume of water ejected

A = area of cross section of bore = 10 cm^2

ρ = density of water = 1 g/cc

l = length of the water ejected in time t

$\frac{dl}{dt} = v =$ velocity of water ejected

= 0.5 m/s = 50 cm/s

$$F = \frac{dm}{dt}v = \left(A\rho v\right)v = A\rho v^2 = 10\times1\times50^2$$

$$= 25000 \text{ dyne} = 0.25 \text{ N}$$

Equal and opposite force must be applied by the gardener.

4.4. Inertial and Non-Inertial Frames of Reference

Consider yourself standing on a railway platform or a bus stand and you see a train or bus moving. According to you, that train or bus is moving or is in motion. As per the experience of the passengers in the train or bus, they are at rest and you are moving (in backward direction). Hence motion itself is a relative concept. To know or describe a motion you need to describe or define some reference. Such a reference is called a *frame of reference*. In the example discussed above, if you consider the platform as the reference, then the passengers and the train are moving. However, if the train is considered as the reference, you and platform, etc. are moving.

Usually a set of coordinates with a suitable origin is enough to describe a frame of reference. If position coordinates of an object are continuously changing with time in a frame of reference, then *that object* is in motion in *that* frame of reference. Any frame of reference in which Newton's first law of motion is applicable is the simplest understanding of an

inertial frame of reference. It means, if there is no net force, there is no acceleration. Thus in an inertial frame, a body will move with constant velocity (which may be zero also) if there is no net force acting upon it. In the absence of a net force, if an object suffers an acceleration, that frame of reference is ***not*** an inertial frame and is called as non-inertial frame of reference.

Measurements in one inertial frame can be converted to measurements in another inertial frame by a simple transformation, i.e., by simply using some velocity vectors (relative velocity between the two frames of reference).

Illustration: Imagine yourself inside a car with all windows opaque so that you can not see anything outside. Also consider that there is a pendulum tied inside the car and not set into oscillations . If the car just starts its motion (with reference to outside or ground), you will experience a jerk, i.e., acceleration inside the car even though there is no force acting upon you. During this time, the string of the pendulum may be steady, but ***not*** vertical. During time of acceleration, the car can be considered to be a non-inertial frame of reference. Later on if the car is moving with constant velocity (with reference to the ground), you will not experience any jerky motion within the car and the car can be considered as an inertial frame of reference. In this case, the pendulum string will be vertical, when not oscillating.



Do you know ?

The situations/phenomena that can be explained using Newton's laws of motion fall under Newtonian mechanics. So far as our daily life situations are considered, Newtonian mechanics is perfectly applicable. However, under several extreme conditions we need to use some other theories.

Limitation of Newton's laws of motion

(i) Newton's laws are applicable only in the inertial frames of reference (discussed later). If the body is in a frame of reference of acceleration (a), we need to use a *pseudo* force $\left(-m\vec{a}\right)$ in addition to all the other forces while writing the force equations.

(ii) Newton's laws are applicable for *point* objects.

(iii) Newton's laws are applicable to rigid bodies. A body is said to be *rigid* if the relative distances between its particles do not change for any deforming force.

(iv) For objects moving with speeds comparable to that of light, Newton's laws of motion do not give results that match with the experimental results and Einstein's special theory of relativity has to be used.

(v) Behaviour and interaction of objects having atomic or molecular sizes cannot be explained using Newton's laws of motion, and quantum mechanics has to be used.

A rocket in intergalactic space (gravity free space between galaxies) with all its engines shut is closest to an ideal inertial frame. However, Earth's acceleration in the reference frame of the Sun is so small that any frame attached to the Earth can be used as an inertial frame for any day-to-day situation or in our laboratories.

4.5 Types of Forces:

4.5.1. Fundamental Forces in Nature:

All the forces in nature are classified into following **four** interactions that are termed as fundamental forces.

(i) Gravitational force: It is the attractive force between two (point) masses separated by a distance. Magnitude of gravitational force between point masses m_1 and m_2 separated by distance r is given by $F = \frac{Gm_1m_2}{r^2}$

where G = 6.67×10^{-11} SI units. Between two point masses (particles) separated by a given distance, this is the weakest force having infinite range. This force is always attractive. Structure of the universe is governed by this force.

Common experience of this force for us is gravitational force exerted by Earth on us, which we call as our weight W.

$$\therefore W = \frac{GMm}{R^2} = m\left(\frac{GM}{R^2}\right) = mg$$,

where M and R represent respectivily mass and radius of the Earth. Distance between ourselves and Earth is taken as radius of the Earth when we are on the surface of the Earth because our size is negligible as compared to radius of the Earth (6.4×10^6 m).

$$\left(\frac{GM}{R^2}\right) = \frac{6.67\times10^{-11}\times6\times10^{24}}{\left(6.4\times10^6\right)^2}$$

$\cong 9.8$ m/s^2 = g = gravitational acceleration or gravitational field intensity.

We ***feel*** this force only due to normal reaction from the surface of our contact with Earth.

All individual bodies also exert gravitation force on each other but it is too small compared to that by the Earth. For example, mutual gravitational force between two SUMO wrestlers, each of mass 300 kg, assuming the distance between them is 0.5 m, will be

$$F = \frac{6.67\times10^{-11}\times300\times300}{0.5^2}$$

$\cong 24\times10^{-5}$ N $= 24\ \mu$N.

This force is negligibly small in comparison to the weight of each SUMO wrestler $\cong$ 3000 N.

(ii) Electromagnetic (EM) force: It is an attractive or repulsive force between electrically charged particles. Earlier, electric and magnetic forces were thought to be independent. After the demonstrations by Michael Faraday (1791-1867) and James Clerk Maxwell (1831-1879), electric and magnetic forces were unified through the theory of electromagnetism. These forces are stronger than the gravitational force. Our life is practically governed by these forces. Majority of forces experienced in our daily life, such as force of friction, normal reaction, tension in strings, collision forces, elastic forces, viscosity (fluid friction), etc. are EM in nature. Under the action of these forces, there is *deformation of objects* that changes intermolecular distances thereby resulting into reaction forces.

(iii) Strong (nuclear) force: This is the strongest force that binds the nucleons together inside a nucleus. Though strongest, it is a short range ($< 10^{-14}$ m) force. Therefore is very strong attractive force and is charge independent.

(iv) Weak (nuclear) force: This is the interaction between subatomic particles that is responsible for the radioactive decay of atoms, in particular beta emission. The weak nuclear force is not as weak as the gravitational force, but much weaker than the strong nuclear and EM forces. The range of weak nuclear force is exceedingly small, of the order of 10^{-16} m.

Weak interaction force:

The radioactive isotope C^{13} is converted into N^{14} in which a neutron is converted into a proton. This property is used in *carbon dating* to determine the age of a sample.

In radioactive beta decay, the nucleus emits an electron (or positron) and an uncharged particle called neutrino. There are two types of β-decay, β^+ and β^-. During β^+decay, a proton is converted into a neutron (accompanied by positron emission) and during β^- decay a neutron is converted into a proton (accompanied by electron emission).

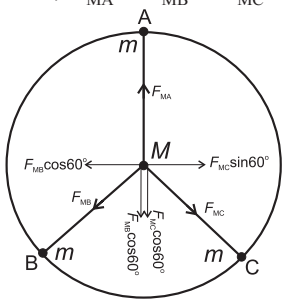
Another most interesting illustration of weak forces is fusion reaction in the core of the Sun. During this, protons are converted into neutrons and a neutrino is emitted due to energy balance. In general, emission of a neutrino is the evidence that there is conversion of a proton into a neutron or a neutron into a proton. This is possible *only* due to weak forces.

Example 4.2. Three identical point masses are fixed symmetrically on the periphery of a circle. Obtain the resultant gravitational force on any point mass M at the centre of the circle. Extend this idea to more than three identical masses symmetrically located on the periphery. How far can you extend this concept?

Solution:

(i) Figure below shows three identical point masses m on the periphery of a circle of radius r. Mass M is at the centre of the circle. Gravitational forces on M due to these masses are attractive and are directed as shown.

In magnitude, $F_{MA} = F_{MB} = F_{MC} = \dfrac{GMm}{r^2}$



Forces F_{MB} and F_{MC} are resolved along F_{MA} and perpendicular to F_{MA} as shown. Components perpendicular to F_{MA} cancel each other. Components along F_{MA} are $F_{MB}\cos 60^0 = F_{MC}\cos 60^0 = \dfrac{1}{2}F_{MA}$ each. Magnitude of their resultant is F_{MA} and its direction is opposite to that of F_{MA}. Thus, the resultant force on mass M is zero.

(ii) For any even number of equal masses, the force due to any mass m is balanced (cancelled) by diametrically opposite mass. For any odd number of masses, as seen for 3, the components perpendicular to one of them cancel each other while the components parallel to one of these add up in such a way that the resultant is zero for any number of identical masses m located symmetrically on the periphery.

(iii) As the number of masses tends to infinity, their collective shape approaches circumference of the circle, which is nothing but a ring. Thus, the gravitational force exerted by a ring mass on any other mass at its centre is zero.

In three-dimensions, we can imagine a uniform hollow sphere to be made up of infinite number of such rings with a common diameter. Thus, the gravitational force for any mass kept at the centre of a hollow sphere is zero.

Do you know ?

Unification of forces: Newton unified terrestrial (related to Earth and hence to our daily life) and celestial (related to universe) domains under a common law of gravitation. The experimental discoveries of Oersted (1777-1851) and Faraday showed that electric and magnetic phenomena are in general inseparable leading to what is called 'EM phenomenon'. Electromagnetism and optics were unified by Maxwell with the proposition that light is an EM wave. Einstein attempted to unify gravity and electromagnetism under general relativity but could not succeed. The EM and the weak nuclear force have now been unified as a single 'electro-weak' force.

4.5.2. Contact and Non-Contact Forces:

For some forces, like gravitational force, electrostatic force, magnetostatic force, etc., physical contact is not an essential condition. These forces exist even if the objects are distant or physically separated. Such forces are non-contact forces.

Forces resulting *only* due to contact are called contact forces. All these are EM in nature, arising due to some deformation. Normal reaction, forces occurring during collision, force of friction, etc., are contact forces. There are two common categories of contact forces. Two objects in contact, while exerting mutual force, try to push each other away along their common normal. Quite often we call it as 'normal reaction' force or 'normal' force. While standing on a table, we *push* the table away from us (downward) and the table *pushes* us away from it (upward) both being equal in magnitude and acting along the same '*normal*' line.

Force of friction is also a contact force that arises whenever there is a relative motion or tendency of relative motion between surfaces in contact. This is the parallel (or tangential) component of the reaction force. In this case, the molecules of surfaces in contact, which have developed certain equilibrium, are required to be separated.

4.5.3 Real and Pseudo Forces:

Consider ourselves inside a lift (or elevator). When the lift just starts moving up (accelerates upward), we feel a bit heavier as if someone is pushing us down. This is not imaginary or not just a feeling. If we are standing on a weighing scale inside this lift, during this time the weighing scale records an increase in weight. During travelling with uniform upward velocity no such change is recorded. While stopping at some upper floor, the lift undergoes downward acceleration for decreasing the upward velocity. In this case the weighing scale records loss in weight and we also feel lighter. These *extra* upward or downward forces are (i) Measurable, means they are not imaginary, (ii) not accountable as per Newton's second law in the inertial frame and (iii) not among any of the four fundamental forces.

When we are inside a bus such forces are experienced when the bus starts to move (forward acceleration), when the bus is about to stop (backward acceleration) or takes a turn (centripetal acceleration). In all these cases we are inside an accelerated system (which is our frame of reference). If a force measuring device is suitably used – like the weighing scale recording the change in weight – these forces can be recorded and will be found to be always opposite to the acceleration of your frame of reference. They are also exactly equal to $-m\vec{a}$, where m is our mass and $\vec{a}$ is acceleration of the system (frame of reference).

We have already defined or described real forces to be those which obey Newton's laws of motion and are one of the four fundamental forces. Forces in above illustrations do not satisfy this description and cannot be called real forces. Hence these are called ***pseudo*** forces. ***Pseudo*** in this case does not mean imaginary (because these are measurable with instruments) but just means *non-real.* These forces are measured to be $(-m\vec{a})$. Hence, a term $(-m\vec{a})$ added to resultant force enables us to apply Newton's laws of motion to a non-inertial frame of reference of acceleration $\vec{a}$. Negative sign refers to their direction, which is opposite to that of the acceleration of the reference frame.

As per the illustration of the lift with downward acceleration $\vec{a}$, the weight experienced will be $\vec{W} = m\vec{g} + (-m\vec{a})$

As $\vec{g}$ and $\vec{a}$ are along the same direction in this case, $W = mg - ma$. This explains the *feeling* of a loss in weight.

During upward acceleration, say $\vec{a_1}$, we have, $\vec{W_1} = m\vec{g} + (+m\vec{a_1})$

In this case, $\vec{g}$ and $\vec{a_1}$ are oppositely directed. $\therefore W_1 = mg + (+ma_1) = mg + ma_1$ that explains gain in weight or existence of extra downward force.

> In mathematics we define a number to be real if its square is zero or positive. Solution set of equations like $x^2 - 6x + 10 = 0$ does not satisfy the criterion to be a real number. Such numbers are *complex* numbers which include $i = \sqrt{-1}$ along with some real part. It means every *non-real* need not be *imaginary* as in literal verbal sense.

Example 4.3: A car of mass 1.5 ton is running at 72 kmph on a straight horizontal road. On turning the engine off, it stops in 20 seconds. While running at the same speed, on the same road, the driver observes an accident 50 m in front of him. He immediately applies the brakes and just manages to stop the car at the accident spot. Calculate the braking force.

Solution: On turning the engine off,

$u = 20\,\text{m s}^{-1}$, $v = 0$, $t = 20\,s$

$$\therefore a = \frac{v - u}{t} = -1\,\text{m s}^{-2}$$

This is frictional retardation (negative acceleration).

After seeing the accident,

$u = 20\,\text{m s}^{-1}$, v = 0, $s = 50\,\text{m}$

$$\therefore a_1 = \frac{v^2 - u^2}{2s} = -4\,\text{m s}^{-2}$$

This retardation is the combined effect of braking and friction. Thus, braking retardation $= 4 - 1 = 3\,\text{m s}^{-2}$.

$\therefore$ Braking force = mass × braking retardation = 1500 × 3 = 4500 N

4.5.4. Conservative and Non-Conservative Forces and Concept of Potential Energy:

Consider an object lying on the ground is lifted and kept on a table. Neglecting air resistance, the amount of work done is the work done *against* gravitational force and it is *independent* of the actual path chosen (Remember, as there is no air resistance, gravitational force is the only force). Similarly, while keeping the same object back on the ground from the table, the work is done ***by*** the gravitational force. In either case the amount of work done is the same ***and*** is independent of the actual path chosen. The work done by force $\vec{F}$ in moving the object through a distance dx can be mathematically represented as $dW = \vec{F}.d\vec{x}$ $= -dU$ or $dU = -\vec{F}.d\vec{x}$.

If work done by or against a force is independent of the actual path, the force is said to be a ***conservative force***. During the work done by a conservative force, the mechanical energy (sum of kinetic and potential energy) is conserved. In fact, we define the concept of potential at a point or potential energy (in the topic of gravitation) with the help of conservative forces only. The work done by or against conservative forces reflects an equal amount of change in the *potential energy*. The corresponding work done is used in changing the position or in achieving the new position in the gravitational field. Hence, potential energy is often referred to as the energy possessed on account of position.

In the illustration given above, the work done is reflected as increase in the gravitational potential energy when the displacement is *against* the (gravitational) force. Same amount of potential energy is decreased when the displacement is in the direction of force. In either case it is independent of the actual path but depends only upon the initial and final positions. This change in the potential energy takes place in such a way that the mechanical energy is conserved.

As discussed above, increase in the potential energy, $dU = -\vec{F}.d\vec{x}$ or $U = -\int \vec{F}.d\vec{x}$ where $\vec{F}$ is a conservative force. This concept, will be described in details in Chapter 5 on Gravitation in context of gravitational potential energy and gravitational potential.

During this process, if friction or air resistance is present, additional work is necessary against the frictional force (for the same displacement). This work is strictly path dependent and not recoverable. Such forces (like friction, air drag, etc.) are called *non-conservative* forces. Work done against these forces appears as heat, sound, light, etc. The work done against non-conservative forces is not recoverable even if the path is exactly reversed.

4.5.5. Work Done by a Variable Force:

The popular formula for calculating work done is $W = \vec{F} \cdot \vec{s} = F\,s\cos\theta$ where θ is the angle between the applied force $\vec{F}$ and the displacement $\vec{s}$.

This formula is applicable only if both force $\vec{F}$ and displacement $\vec{s}$ are constant and finite. In several real-life situations, the force is not constant. For example, while lifting an object through several thousand kilometres, the gravitational force is not constant. The viscous forces like fluid resistance depend upon the speed, hence, quite often are not constant with time. In order to calculate the work done by such variable forces we use integration.

Illustration: Figure 4.1(a) shows variation of a force $\vec{F}$ plotted against corresponding displacements in its direction $\vec{s}$. As the displacement is in the direction of the applied force, vector nature is not used. We need to calculate the work done by this force during

displacement from s_1 to s_2. As the force is variable, using $W = F(s_2 - s_1)$ directly is not possible. In order to use integration, let us divide the displacement into a large number of infinitesimal (infinitely small) displacements. One of such displacements is *ds*. It is so small that the force *F* is practically constant for this displacement. Practically constant means the change in the force is so small that the change can not be recorded. The shaded strip shows one of such displacements. As the force is constant, the area of this strip *F.ds* is the work done *dW* for this displacement. Total work done *W* for displacement $(s_2 - s_1)$ can then be obtained by using integration.

$$\therefore W = \int_{s_1}^{s_2} F.ds$$

Method of integration is applicable if the exact way of variation in $\vec{F}$ with $\vec{s}$ is known and that function is integrable.

The area under the curve between s_1 and s_2 also gives the work done *W* (if the force axis necessarily starts with zero), as it consists of all the strips of *ds* between s_1 and s_2. In Fig. 4.1(b), the variation in the force is linear. In this case, the area of the trapezium AS_1S_2B gives total work done *W*.

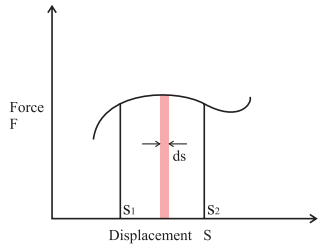


Fig 4.1 (a): Work done by nonlinearly varying force.

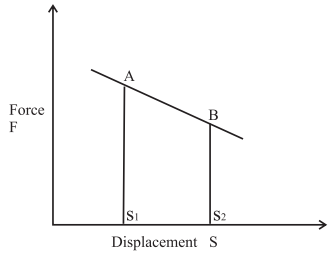


Fig 4.1 (b): Work done by linearly varying force.

Example 4.4: Over a given region, a force (in newton) varies as $F = 3x^2 - 2x + 1$. In this region, an object is displaced from $x_1 = 20$ cm to $x_2 = 40$ cm by the given force. Calculate the amount of work done.

Solution:

$$W = \int_{s_1}^{s_2} F.ds = \int_{0.2}^{0.4} (3x^2 - 2x + 1)dx$$

$$= \left[x^3 - x^2 + x\right]_{0.2}^{0.4}$$

$$= \left[0.4^3 - 0.4^2 + 0.4\right] - \left[0.2^3 - 0.2^2 + 0.2\right]$$

$$= 0.304 - 0.168 = 0.136\,\text{J}$$

4.6. Work Energy Theorem:

If there is a decrease in the potential energy (like a body falling down) due to a conservative force, it is entirely converted into kinetic energy. Work done by the force then appears as kinetic energy. Vice versa if an object is moving against a conservative force its kinetic energy decreases by an amount equal to the work done against the force. This principle is called *work-energy theorem* for conservative forces.

Case I: Consider an object of mass *m* moving with velocity $\vec{u}$ experiencing a constant opposing force $\vec{F}$ which slows it down to $\vec{v}$ during displacement $\vec{s}$. As $\vec{u}$ and $\vec{F}$ are oppositely directed, the entire motion will be along the same *line*. In this case we need not use the vector form, just ± signs should be good enough.

If $a = \frac{F}{m}$ is the acceleration, we can write the relevant equation of motion as $v^2 - u^2 = 2(-a)s$ (negative acceleration for opposing force)

Multiplying throughout by m/2, we get

$$\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = (ma).s = F.s$$

Left-hand side is decrease in the kinetic energy and the right-hand side is the work done by the force. Thus, change in kinetic energy is equal to work done by the conservative force, which is in accordance with work-energy theorem.

Case II: Accelerating conservative force along

with a retarding non-conservative force:

An object dropped from some point at height h falls down through air. While coming down its potential energy decreases. Equal amount of work is done in this case also. However, this time the work is not entirely converted into kinetic energy but some part of it is used in overcoming the air resistance. This part of energy appears in some other forms such as heat, sound, etc. In this case, the work-energy theorem can mathematically be written as $\Delta PE = \Delta KE + W_{air\ resistance}$

(Decrease in the gravitational P.E. = Increase in the kinetic energy + work done against non-conservative forces). Magnitude of air resistance force is not constant but depends upon the speed hence it can be written as $\int \vec{F}.\vec{ds}$ as seen during work done by (or against) a variable force.

4.7. Principle of Conservation of Linear Momentum:

According to Newton's second law, resultant force is equal to the rate of change of linear momentum or $\vec{F} = \frac{d\vec{p}}{dt}$

In other words, if there is no resultant force, the linear momentum will not change or will remain constant or will be conserved. Mathematically, if $\vec{F} = 0, \frac{d\vec{p}}{dt} = 0$ or $\vec{p}$ is constant

Always remember

Isolated system means absence of any external force. A *system* refers to a set of particles, colliding objects, exploding objects, etc. *Interaction* refers to collision, explosion, etc. During any interaction among such objects the *total* linear momentum of the *entire system* of these particles/objects is constant. Remember, forces during collision or during explosion are *internal* forces for that *entire system*.

During collision of two particles, the two particles exert forces on *each other*. If these particles are discussed independently, these are external forces. However, for the *system of the two particles* together, these forces are internal forces.

(or conserved). This leads us to the *principle of conservation of linear momentum* which can be stated as **"The total momentum of an isolated system is conserved during any interaction."**

Systems and free body diagrams:

Mathematical approach for application of Newton's second law:

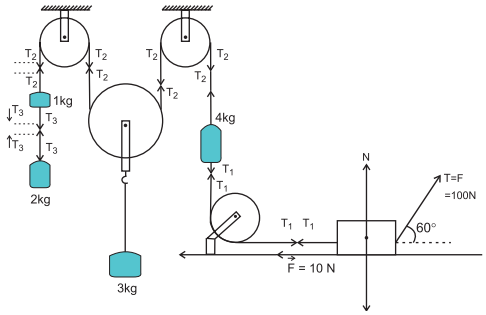


Fig 4.2 (a): System for illustration of free body diagram.

Consider the arrangement shown in Fig.4.2 (a). Pulleys P_1, P_3 and P_4 are fixed, while P_2 is movable. Force F = 100 N, applied at an angle 60° with the horizontal is responsible for the motion, if any. Contact surface of the 5 kg mass offers a constant opposing force F = 10 N. Except this, there are no resistive forces anywhere.

Discussion: Until 1 kg mass reaches the pulley P_1, the motion of 1 kg and 2 kg masses is identical. Thus, these two can be considered to be a *single system* of mass 3 kg except for knowing the tension T_3. The forces due to tension in the string joining them are internal forces for this system.

All masses *except* the 3 kg mass are travelling same distance in the same time. Thus, their accelerations, if any, have same magnitudes. If the string S connecting 1 kg and 4 kg masses moves by x, the lower string S_1 holding the 3 kg mass moves through $x/2$.

Free body diagrams (FBD): A free body diagram refers to forces acting on only one body at a time, and its acceleration.

Free body diagram of 2 kg mass: Let a be its upward acceleration. According to Newton's second law, it must be due to the resultant vertical force on this mass. To know this force,

we need to know all the individual forces acting on this mass. The agencies exerting forces on this mass are Earth (downward force 1g) and force due to the tension T_3.

In this case, the lower half of the string is connected to the 2 kg mass. The direction of T_3 for lower part of the string is upwards as shown in the Fig. 4.2 (b). Upper part of the string is connected to the 1 kg mass. Thus, the direction of T_3 for 1 kg mass will be downwards. However, it will appear only for the free body diagram of the 1 kg mass and will not appear in the free body diagram of 2 kg mass. Hence, the free body diagram of the 2 kg mass will be as follows: Its force equation, according to Newton's second law will then be $T_3 - 2g = 2a$.

Practical tip: Easiest way to know the direction of forces due to tension is to put an X-mark on the string. Two halves of this cross indicate the directions of the forces exerted *by the string* on the bodies connected to either parts of the string.

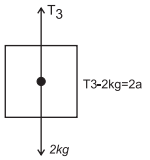


Fig 4.2 (b): Free body diagram for 2 kg mass.

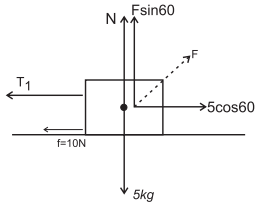


Fig 4.2 (c): Free body diagram for 5 kg mass.

Free body diagram of mass 5 kg: Its horizontal acceleration is also *a*, but towards right. The force exerting agencies are Earth (force 5g downwards), contact surface (normal force *N,* vertically upwards ***and*** opposing force F = 10 N, towards left), and the two strings on either side (Forces due to their tensions T and T_1). All these are shown in its free body diagrams in Fig. 4.2 (c). On resolving the force F along the vertical and horizontal directions, the free body diagram of 5 kg mass can be drawn as explained below.

As this mass has only horizontal motion, the vertical forces must cancel. Therefore, along the vertical direction,

$$N + F \sin 60° = 5g$$

Along the horizontal direction,

$$T_1 + 10(\text{opposing force}) = F\cos 60^0 = T\cos 60^0$$

Similar equations can be written for all the masses and also for the movable pully. On solving these equations simultaneously, we can obtain all the necessary quantities.

Example 4.5: Figure (a) shows a fixed pulley. A massless inextensible string with masses m_1 and $m_2 > m_1$ attached to its two ends is passing over the pulley. Such an arrangement is called an Atwood machine. Calculate accelerations of the masses and force due to the tension along the string assuming axle of the pulley to be frictionless.

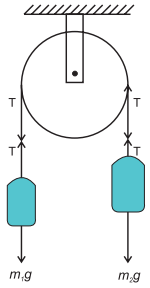


Fig. (a).

Solution: **Method I:** Direct method: As $m_2 > m_1$, mass m_2 is moving downwards and mass m_1 is moving upwards.

Net downward force

$$= F = (m_2)g - (m_1)g = (m_2 - m_1)g$$

As the string is inextensible, both the masses travel the same distance in the same time. Thus, their accelerations are numerically the same (one upward, other downward). Let it be *a*.

Thus, total mass in motion, $M = m_2 + m_1$

$$\therefore a = \frac{F}{M} = \left(\frac{m_2 - m_1}{m_2 + m_1}\right)g$$

For mass m_1, the upward force is the force due to tension T and downward force is mg. It has upward acceleration *a.* Thus, $T - m_1 g = ma$

$\therefore T = m_1(g + a)$

Using the expression for *a*, we get

$$T = \left(\frac{2m_1m_2}{m_1+m_2}\right)g$$

Method II: (Free body equations)

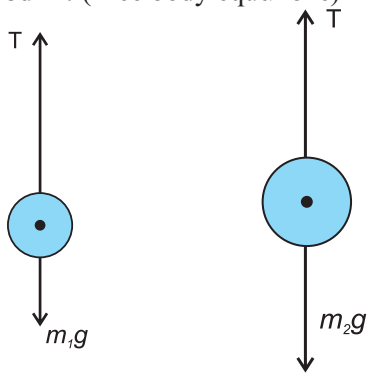


Fig. (b) Fig. (b)

Free body diagrams of m_1 and m_2 are as shown in Figs. (b) and (c).

Thus, for the first body, $T - m_1g = m_1a$ --- (I)

For the second body, $m_2g - T = m_2a$ --- (II)

Adding (I) and (II), and solving for a,

we get, $a = \left(\frac{m_2 - m_1}{m_2 + m_1}\right)g$ --- (III)

Solving Eqs. (II) and (III) for T, we get,

$$T = m_2(g - a) = \left(\frac{2m_1m_2}{m_1+m_2}\right)g$$

4.8. Collisions:

During collisions a number of objects come together, interact (exert forces on *each other*) and scatter in different directions.

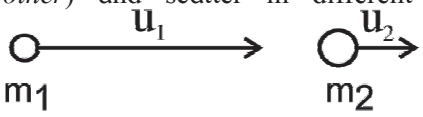


Fig. 4.3 (a): Head on collision-before collision.

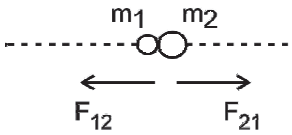


Fig. 4.3 (b): Head on collision-during impact.

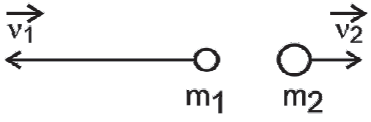


Fig. 4.3 (c): Head on collision-after collision.

4.8.1. Elastic and inelastic collisions:

During a collision, the linear momentum of the entire system of particles is always conserved as there is no external force acting on the *system* of particles. However, the individual momenta of the particles change due to mutual forces, which are internal forces.

$\therefore \sum\vec{p}_{initial} = \sum\vec{p}_{final}$, during ***any*** collision (or explosion), where $\vec{p}$'s are the linear momenta of the particles.

However, kinetic energy of the entire system may or may not conserve.

Collisions can be of two types: ***elastic collisions*** and ***inelastic collisions.***

Elastic collision: A collision is said to ***elastic*** if kinetic energy of the entire system is conserved during the collision (along with the linear momentum). Thus, during an elastic collision,

$$\sum K.E._{initial} = \sum K.E._{final}$$

An elastic collision is *impossible* in daily life. However, in many situations, the interatomic or intermolecular collisions are considered to be elastic (like in kinetic theory of gases, to be discussed in the next standard).

Inelastic Collision: A collision is said to be ***inelastic*** if there is a loss in the kinetic energy during collision, but linear momentum is conserved. The loss in kinetic energy is either due to internal friction or vibrational motion of atoms causing heating effect. Thus, during an inelastic collision,

$$\sum K.E._{initial} > \sum K.E._{final}.$$

During an explosion as energy is supplied internally. Thus,

$$\sum K.E._{final} > \sum K.E._{initial}.$$

As stated earlier, $\sum\vec{p}_{initial} = \sum\vec{p}_{final}$ for inelastic collisions or explosion also. In fact, this is always the first equation for discussing these interactions or while solving numerical questions.

4.8.2. Perfectly Inelastic Collision:

This is a special case of inelastic collisions. If colliding bodies join together after collision, it is said to be a ***perfectly inelastic collision***.

In other words, the colliding bodies have a common final velocity after a perfectly inelastic collision. Being an inelastic collision, obviously there is a loss in the kinetic energy of the system during a perfectly inelastic collision. In fact, the loss in kinetic energy is maximum in perfectly inelastic collision.

Illustrations:

(i) Consider a bullet fired towards a block kept on a smooth surface. Collision between bullet and the block will be elastic if the bullet rebounds with exactly the same initial speed and the block remains stationory. If the bullet gets embedded into the block and the two move jointly, it is perfectly inelastic collision. If the bullet rebounds with smaller speed or comes out of the block on the other side with some speed, it is an inelastic (or partially inelastic) collision. Remember, there is nothing called a partially elastic collision Elastic collisions are always perfectly elastic. An inelastic collision however, may be partially or perfectly inelastic.

(ii) Visualise a ball dropped from some height on a hard surface, the entire system being in an evacuated space. If the ball rebounds exactly to the same height from where it was dropped, the collision between the ball and the surface (in turn, with the Earth) is elastic. As you know, the ball never reaches the same initial height or a height greater than the initial height. Rebounding to smaller height refers to inelastic collision. Instead of ball, if mud or clay is dropped, it sticks to the surface. This is perfectly inelastic collision.

4.8.3. Coefficient of Restitution *e*:

For collision of two objects, the negative of ratio of relative velocity of separation to relative velocity of approach is defined as the coefficient of restitution e.

One dimensional or head-on collision: A collision is said to be head-on if the colliding objects move along the same straight line, before and after the collision. Here, we use u_1, u_2, v_1, v_2 as symbols.

Consider such a head-on collision of two bodies of masses m_1 and m_2 with respective initial velocities u_1 and u_2. As the collision is head on, the colliding masses are along the same line before and after the collision. Hence, vector treatment is not necessary. (**However, velocities must be substituted with proper signs in actual calculation**). Relative velocity of approach is then $u_a = u_2 - u_1$

Let v_1 and v_2 be their respective velocities after the collision. The relative velocity of recede (or separation) is then $v_s = v_2 - v_1$

$$\therefore \text{Coefficient of restitution, } e = -\frac{v_s}{u_a}$$

$$= \frac{-)v_2 - v_1(}{u_2 - u_1} = \frac{v_1 - v_2}{u_2 - u_1}$$

$$= \frac{\text{Relative speed of separation}}{\text{Relative speed of approach}} \qquad \text{---(4.1)}$$

For a perfectly inelastic collision, the colliding bodies move jointly after the collision, i.e., $v_2 = v_1$ or $v_2 - v_1 = 0$. Hence, for a perfectly inelastic collision, $e = 0$. In other words, if $e = 0$, the head-on collision is perfectly inelastic collision.

Coefficient of restitution during a head-on, elastic collision:

Consider the collision described above to be elastic. According to the principle of conservation of linear momentum,

Total initial momentum = Total final momentum.

$$\therefore m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \qquad \text{--- (4.2)}$$

$$\therefore m_1(u_1 - v_1) = m_2(v_2 - u_2) \qquad \text{--- (4.3)}$$

As the collision is elastic, total kinetic energy of the system is also conserved.

$$\therefore \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \text{ --- (4.4)}$$

$$\therefore m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2)$$

$$\therefore m_1(u_1 + v_1)(u_1 - v_1)$$

$$= m_2(u_2 + v_2)(v_2 - u_2) \qquad \text{--- (4.5)}$$

Dividing Eq. (4.5) by Eq. (4.3), we get

$$u_1 + v_1 = u_2 + v_2 \qquad \text{--- (4.6)}$$

$$\therefore u_2 - u_1 = v_1 - v_2$$

For an elastic collision,

$$e = \frac{v_1 - v_2}{u_2 - u_1} = 1 \qquad \text{--- (4.7)}$$

Thus, for an elastic collision, coefficient of restitution, $e = 1$. For a perfectly inelastic collision, $e = 0$ (by definition). Thus, for any collision, the coefficient of restitution lies between 1 and 0.

Above expressions (Eq. (4.1) Eq. (4.7)) are general. While substituting the values of u_1, u_2, v_1, v_2, their algebraic values must be used in actual calculation.

For example referring to the

Fig. 4.3 (a), (b) and (c)

Eq (4.1) gives

$$e = -\frac{v_s}{u_a}$$

Here $u_a = u_1 - u_2$ since $u_1 > u_2$

and $v_s = v_1 + v_2$ since the objects go in opposite directions.

$$\therefore e = -\frac{v_1 + v_2}{u_1 - u_2} \qquad \text{--- (a)}$$

Using Eq (4.6),

$u_1 + v_1 = u_2 + v_2$

$\therefore$ According to Fig. 4.3,

$u_1 - v_1 = u_2 + v_2$

$\therefore v_1 + v_2 = u_2 - u_1$ --- (b)

By substituting in (a),

$$e = -\frac{(u_2 - u_1)}{(u_1 - u_2)} = 1,$$

which is the case of a perfectly elastic collision.

4.8.4. Expressions for final velocities after a head-on, elastic collision:

From Eq. (4.6), $v_2 = u_1 + v_1 - u_2$

Using this in Eq. (4.2), we get

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2(u_1 + v_1 - u_2)$$

$$\therefore v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2 \qquad \text{--- (4.8)}$$

Subscripts 1 and 2 were arbitrarily chosen. Thus, just interchanging 1 with 2 gives us v_2 as

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \left(\frac{2m_1}{m_1 + m_2}\right)u_1 \qquad \text{--- (4.9)}$$

Equation (4.9) can also be obtained by substituting v_1 from Eq. (4.8) in Eq. (4.6).

Particular cases:

(i) If the bodies are of equal masses (or identical), $m_1 = m_2$, Eqs. (4.8) and (4.9) give

$v_1 = u_2$ and $v_2 = u_1$.

Thus, the bodies just exchange their velocities.

(ii) If colliding body is much heavier and the struck body is initially at rest, i.e.,

$$m_1 \gg m_2 \text{ and } u_2 = 0,$$

we can use

$$m_1 \pm m_2 \cong m_1 \text{ and } \frac{m_2}{m_1 + m_2} \cong 0$$

$\therefore v_1 \cong u_1$ and $v_2 \cong 2u_1$, i.e., the massive striking body is practically unaffected and the tiny body which is struck, travels with double the speed of the massive striking body.

(iii) The body which is struck is much heavier than the colliding body and is initially at rest, i.e., $m_1 \ll m_2$ and $u_2 = 0$.

Using similar approximations, we get, $v_1 \cong -u_1$ and $v_2 \cong 0$, i.e., the tiny (lighter) object rebounds with same speed while the massive object is unaffected. This is as good as dropping an elastic object on hard surface of the Earth .



Do you know ?

Are you aware of elasticity of materials? Is there any connection between elasticity of materials and elastic collisions?

Example 4.6: One marble collides head-on with another identical marble at rest. If the collision is partially inelastic, determine the ratio of their final velocities in terms of coefficient of restitution e.

Solution: According to conservation of momentum, $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

As $m_1 = m_2$, we get, $u_1 + u_2 = v_1 + v_2$

$\therefore$ If $u_2 = 0$, we get, $v_1 + v_2 = u_1$ ... (I)

Coefficient of restitution,

$$e = \frac{v_2 - v_1}{u_1 - u_2} \quad \therefore v_2 - v_1 = eu_1 \text{ ... (II)}$$

Dividing Eq. (I) by Eq. (II),

$$\frac{v_1 + v_2}{v_2 - v_1} = \frac{1}{e}$$

Using componendo and dividendo, we get,

$$\frac{v_2}{v_1} = \frac{1+e}{1-e}$$

4.8.5. Loss in the kinetic energy during a perfectly inelastic head-on collision:

Consider a perfectly inelastic, head on collision of two bodies of masses m_1 and m_2 with respective initial velocities u_1 and u_2. As the collision is perfectly inelastic, they move jointly after the collision, i.e., their final velocity is the same. Let it be v.

According to conservation of linear momentum, $m_1u_1 + m_2u_2 = (m_1 + m_2)v$

$$\therefore v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2} \quad \text{--- (4.10)}$$

This is the common velocity after a perfectly inelastic collision

Loss in K.E. = Δ (K.E.)

= Total initial K.E. - Total final K.E.

$$\therefore \Delta(\text{K.E.}) = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}(m_1 + m_2)v^2$$

$$= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}(m_1 + m_2)\left(\frac{m_1u_1 + m_2u_2}{m_1 + m_2}\right)^2$$

$$\therefore \Delta(\text{K.E.}) = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}\frac{(m_1u_1 + m_2u_2)^2}{(m_1 + m_2)}$$

On simplifying, we get,

$$\Delta(\text{K.E.}) = \frac{1}{2}\left(\frac{m_1m_2}{m_1 + m_2}\right)(u_1 - u_2)^2 \quad \text{--- (4.11)}$$

Masses are always positive and $(u_1 - u_2)^2$ is also positive. Hence, there is always a loss in the kinetic energy in a perfectly inelastic collision.

Final velocities and loss in K.E. in an inelastic head-on collision:

If e is the coefficient of restitution, using Eq. (4.2), the expressions for final velocities after an inelastic collision can be derived as

$$v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2}\right)u_1 + \left(\frac{[1+e]m_2}{m_1 + m_2}\right)u_2$$

$$= \frac{em_2(u_2 - u_1) + m_1u_1 + m_2u_2}{m_1 + m_2}$$ and

$$v_2 = \left(\frac{m_2 - em_1}{m_1 + m_2}\right)u_2 + \left(\frac{[1+e]m_1}{m_1 + m_2}\right)u_1$$

$$= \frac{em_1(u_1 - u_2) + m_1u_1 + m_2u_2}{m_1 + m_2}$$

Loss in the kinetic energy is given by

$$\Delta(\text{K.E.}) = \frac{1}{2}\left(\frac{m_1m_2}{m_1 + m_2}\right)(u_1 - u_2)^2(1 - e^2)$$

.

As $e < 1$, $(1 - e^2)$ is always positive. Thus, there is always a loss of K.E. in an inelastic collision. Also, for a perfectly inelastic collision, $e = 0$. Hence, in this case, the loss is maximum. Using $e = 1$, these equations lead us to an elastic collision and for $e = 0$ they lead us to a perfectly inelastic collision. Verify that they give the same expressions that are derived earlier.

The quantity $\mu = \frac{m_1m_2}{m_1 + m_2}$ is the reduced mass of the system.

Impulse or change in momenta of the bodies:

During collision, the linear momentum delivered by first body (particle) to the second body must be equal to change in momentum or *impulse* of the second body, and vice versa.

$\therefore$ Impulse, $|J| = |\Delta p_1| = |\Delta p_2|$

$$= |m_1v_1 - m_1u_1| = |m_2v_2 - m_2u_2|$$

On substituting the values of v_1 and v_2 and solving, we get

$$|J| = \left(\frac{m_1m_2}{m_1 + m_2}\right)(1 + e)(|u_1 - u_2|)$$

$$= \mu(1 + e)u_{\text{relative}}$$

$|u_1 - u_2| = u_{\text{relative}}$ = velocity of approach

4.8.6. Collision in two dimensions, i.e., a nonhead-on collision:

In this case, the direction of at least one initial velocity is NOT along the line of impact. In order to discuss such collisions mathematically, it is convenient to use two mutually perpendicular directions as shown in Fig. 4.4. One of them is the common tangent at the point of impact, along which there is no force (or along this direction, there is no change in momentum). The other direction is perpendicular to this common tangent through the point of impact, in the two-dimensional plane of initial and final velocities. This is called the line of impact. Internal mutual forces exerted during impact, which are responsible for change in the momenta, are acting along this line. From Fig. (4.4), $\vec{u}_1$ and $\vec{u}_2$, initial velocities make angles α_1 and α_2 respectively with the line of impact while $\vec{v}_1$ and $\vec{v}_2$, final velocities make angles β_1 and β_2 respectively with the line of impact.

According to conservation of linear momentum along the line of contact,

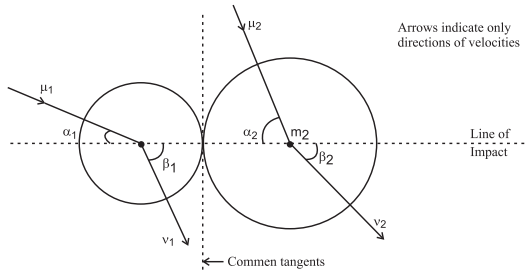


Fig. 4.4: Oblique or non head-on collision.

$$m_1u_1\cos\alpha_1 + m_2u_2\cos\alpha_2 = m_1v_1\cos\beta_1 + m_2v_2\cos\beta_2 \quad \text{--- (4.12)}$$

As there is no force along the common tangent (perpendicular to line of impact),

$$m_1u_1\sin\alpha_1 = m_1v_1\sin\beta_1 \quad \text{--- (4.13)}$$

and $$m_2u_2\sin\alpha_2 = m_2v_2\sin\beta_2 \quad \text{--- (4.14)}$$

For coefficient of restitution, along the line of impact,

$$e = -\left(\frac{v_2\cos\beta_2 - v_1\cos\beta_1}{u_2\cos\alpha_2 - u_1\cos\alpha_1}\right) = \frac{v_2\cos\beta_2 - v_1\cos\beta_1}{u_1\cos\alpha_1 - u_2\cos\alpha_2} \quad \text{--- (4.15)}$$

Equations (4.12), (4.13), (4.14) and (4.15) are to be solved for the four unknowns v_1, v_2, β_1 and β_2

Magnitude of the impulse, along the line of impact,

$$|J| = \left(\frac{m_1m_2}{m_1+m_2}\right)(1+e)\left(|u_1\cos\alpha_1 - u_2\cos\alpha_2|\right)$$

$$= \mu(1+e)u_{\text{relative}}$$

along line of impact.

Loss in the kinetic energy = Δ (K.E.)

$$\frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)(u_1\cos\alpha_1 - u_2\cos\alpha_2)^2(1-e^2)$$

$$= \frac{1}{2}\mu\left(u^2_{\text{relative}}\right)(1-e^2)$$

Example 4.7: A shell of mass 3 kg is dropped from some height. After falling freely for 2 seconds, it explodes into two fragments of masses 2 kg and 1 kg. Kinetic energy provided by the explosion is 300 J. Using g = 10 m/s², calculate velocities of the fragments. Justify your answer if you have more than one options.

Solution: $m_1 + m_2 = 3$ kg.

After falling freely for 2 seconds,

$$v = u + at = 0 + 10(2) = 20\,m\,s^{-1} = u_1 = u_2$$

According to conservation of linear momentum, $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$$\therefore 3\times 20 = 2v_1 + 1v_2 \quad \therefore v_2 = 60 - 2v_1 \text{ --- (I)}$$

K.E. provided = Final K.E. – Initial K.E.

$$= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{1}{2}(m_1+m_2)u^2$$

$$\therefore \frac{1}{2}2v_1^2 + \frac{1}{2}v_2^2 - \frac{1}{2}3(20)^2 = 300\,\text{J}$$

$$\text{or } 2v_1^2 + v_2^2 = 1800$$

$$\text{or } 2v_1^2 + (60-2v_1)^2 = 1800 \text{ using Eq. (I)}$$

$$\therefore 3600 - 240v_1 + 6v_1^2 = 1800$$

$$\therefore v_1^2 - 40v_1 + 300 = 0$$

$$\therefore v_1 = 30\,m\,s^{-1} \text{ or } 10\,m\,s^{-1} \text{ and}$$

$$v_2 = 0 \text{ or } 40\,m\,s^{-1}$$

There are two possible answers since the positions of two fragments can be different as explained below.

If $v_1 = 30$ m s^{-1} and $v_2 = 0$, lighter fragment 2 should be above. On the other hand, if $v_1 = 10$ m s^{-1} and $v_2 = 40$ m s^{-1}, lighter fragment 2 should be below, both moving downwards.

Example 4.8: Bullets of mass 40 g each, are fired from a machine gun at a rate of 5 per second towards a firmly fixed hard surface of area 10 cm^2. Each bullet hits normal to the surface at 400 m/s and rebounds in such a way that the coefficient of restitution for the collision between bullet and the surface is 0.75. Calculate average force and average pressure experienced by the surface due to this firing.

Solution: For the collision,

$u_1 = 400\,\text{m s}^{-1}$, $e = 0.75$, $v_1 = ?$

For the firmly fixed hard surface, $u_2 = v_2 = 0$

$$e = 0.75 = \frac{v_1 - v_2}{u_2 - u_1} = \frac{v_1 - 0}{0 - 400} \quad \therefore v_1 = -300 \text{ m/s.}$$

-ve sign indicates that the bullet rebounds in exactly opposite direction.

Change in momentum of each bullet = m (v_1-u_1)

Equal and opposite will be the momentum transferred to the surface, per collision.

$\therefore$ Momentum transferred to the surface, per collision

$$p = m(u_1 - v_1) = 0.04\big(400 - [-300]\big) = 28 \text{ N s}$$

The rate of collision is same as rate of firing.

$\therefore$ Momentum received by the surface per second, $\frac{dp}{dt} = 28 \times 5 = 140$ N

This must be the *average* force experienced by the surface of area $A = 10$ cm^2 $= 10^{-3}$ m^2

$\therefore$ Average pressure experienced,

$$P = \frac{F}{A} = \frac{140}{10^{-3}} = 1.4 \times 10^5 \text{ N m}^{-2}$$

$\approx$ 1.4 times the atmospheric pressure.

4.9. Impulse of a force:

According to Newton's first law of motion, any unbalanced force changes linear momentum of the system, i.e., basic effect of an unbalanced force is to change the momentum.

According to Newton's second law of motion, $\vec{F} = \frac{d\vec{p}}{dt}$

$$\therefore d\vec{p} = \vec{F}.dt$$

The quantity 'change in momentum' is separately named as *Impulse of the force* $\vec{J}$.

If the force is constant, and is acting for a finite and measurable time, we can write

The change in momentum in time t

$$\vec{J} = d\vec{p} = \vec{p}_2 - \vec{p}_1 = \vec{F}.t \qquad \text{---(4.16)}$$

For a given body of mass m, it becomes

$$\vec{J} = \vec{p}_2 - \vec{p}_1 = m(\vec{v}_2 - \vec{v}_1) = \vec{F}.t \qquad \text{---(4.17)}$$

If $\vec{F}$ is not constant but we know how it varies with time, then

$$\vec{J} = \Delta\vec{p} = \int d\vec{p} = \int \vec{F}.dt \qquad \text{--- (4.18)}$$

Always remember:

1) Colliding objects experience forces along the line of impact which changes their momenta. For their system, these forces are internal forces. These forces form an action-reaction pair, which are equal and opposite, and act on different objects.
2) There is no force along the common tangent, i.e., perpendicular to the line of contact.
3) In reality, the impact is followed by emission of sound and heat and occasionally light. Thus, in general, part of mechanical energy- kinetic energy - is lost (i.e., converted into some other non-recoverable forms). However, total energy of the system is conserved.
4) In reality, velocity of separation (relative final velocity) is less than velocity of approach (relative initial velocity) along the line of impact. Thus coefficient of restitution e < 1.
5) Only during elastic collisions (atomic and molecular level only, never possible in real life), the kinetic energy is conserved and the velocity of separation is equal to the velocity of approach or the initial relative velocity is equal to the final relative velocity.

4.9.1. Necessity of defining impulse:

As discussed above, if a force is constant over a given interval of time or if we know how it varies with time, we can calculate the corresponding change in momentum directly by multiplying the force and time.

However, in many cases, an appreciable force acts for an extremely small interval of

time (too small to measure the force and the time independently). However, change in the momentum due to this force is noticeable and can be measured. This change is defined as *impulse* of the force.

Real life illustrations: While (i) hitting a ball with a bat, (ii) giving a kick to a foot-ball, (iii) hammering a nail, (iv) bouncing a ball from a hard surface, etc., appreciable amount of force is being exerted. In such cases the time for which these forces act on respective objects is negligibly small, mostly not easily recordable. However, the effect of this force is a recordable change in the momentum of that object. Thus, it is convenient to define the change in momentum itself as a physical quantity.

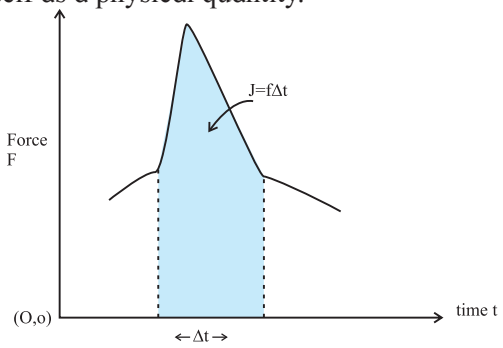


Fig. 4.5: Graphical representation of impulse of a force.

Figure 4.5 shows variation of a force as a function of time e.g., for a collision between bat and ball with the *force axis starting with zero*. The shaded area or the area under the curve gives the product of force against corresponding time (in this case, Δt), hence gives the impulse. For a constant force it is obviously a rectangle. Generally, force is zero before the impact, rises to a maximum and decreases to zero after the impact. For softer tennis ball, the collision time is larger and the maximum force is less. The area under the (F - t) graph is the same. Wicket keeper eases off (by increasing the time of collision) while catching a fast ball. As mentioned earlier, it is absolutely necessary that the force axis must start from zero.

Recall from Chapter 3, analogues concepts using area under a curve are (i) Obtaining displacement in a given time interval as area under the curve for v-t graph, with zero origin for velocity axis. (ii) Obtaining work done by a force as the area under the curve for F-s graph, with zero origin for force axis.

Example 4.9: Mass of an Oxygen molecule is 5.35×10^{-26} kg and that of a Nitrogen molecule is 4.65×10^{-26} kg. During their Brownian motion (random motion) in air, an Oxygen molecule travelling with a velocity of 400 m/s collides elastically with a nitrogen molecule travelling with a velocity of 500 m/s in the exactly opposite direction. Calculate the impulse received by each of them during collision. Assuming that the collision lasts for 1 ms, how much is the average force experienced by each molecule?

Let

$$m_1 = m_O = 5.35 \times 10^{-26} \text{ kg},$$

$$m_2 = m_N = 4.65 \times 10^{-26} \text{ kg}.$$

$$\therefore u_1 = 400 \text{ m s}^{-1} \text{ and } u_2 = -500 \text{ m s}^{-1}$$

taking direction of motion of Oxygen molecule as the positive direction.

For an elastic collision,

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) u_2 \quad \text{and}$$

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2 + \left(\frac{2m_1}{m_1 + m_2}\right) u_1$$

$$\therefore v_1 = -437 \text{ m s}^{-1} \text{ and } v_2 = 463 \text{ m s}^{-1}$$

$$\therefore J_O = m_O (v_1 - u_1) = -4.478 \times 10^{-23} \text{ N s},$$

$$J_N = m_N (v_2 - u_2) = +4.478 \times 10^{-23} \text{ N s}$$

As expected, the net impulse or net change in momentum is zero.

$$F_{ON} = \frac{dp_O}{dt} = \frac{J_O}{\Delta t} = \frac{-4.478 \times 10^{-23}}{10^{-3}}$$

$$= -4.478 \times 10^{-20} \text{ N}$$

and $F_{NO} = -F_{ON} = 4.478 \times 10^{-20}$ N

4.10. Rotational analogue of a force - moment of a force or torque:

While opening a door fixed to a frame on hinges, we apply the force away from the hinges and perpendicular to the door to open it with ease. In this case we are interested in achieving some angular displacement for the door. If the force is applied near the hinges or

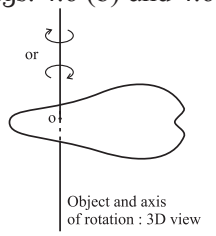
nearly parallel to the door, it is very difficult to open the door. Similarly, if the door is heavier (made up of iron instead of wood or plastic), we need to apply proportionally larger force for the same angular displacement.

It shows that rotational ability of a force not only depends upon the mass (greater force for greater mass), but also upon the point of application of the force (the point should be as away as possible from the axis of rotation) and the angle between direction of the force and the line joining the axis of rotation with the point of application (effect is maximum, if this angle is 90^0).

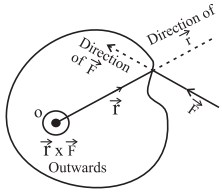
Taking into account all these factors, the quantity **moment of a force** or **torque** is defined as the rotational analogue of a force. As rotation refers to direction (sense of rotation), torque must be a vector quantity. In its mathematical form, torque or moment of a force is given by

$$\vec{\tau} = \vec{r} \times \vec{F} \qquad \text{--- (4.17)}$$

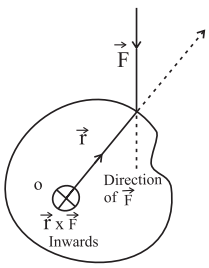
where $\vec{F}$ is the applied force and $\vec{r}$ is the position vector of the point of application of the force from the axis of rotation, as shown in the Figs. 4.6 (b) and 4.6 (c).



Figs. 4.6(a): Illustration of moment of force with object and axis of rotation in 3D view.



Figs. 4.6(b): Top view for moment of force $\vec{F}$ in anticlockwise rotation with $\vec{F}$ and $\vec{r}$ in the plane of paper.



Figs. 4.6(c): Top view of moment of force $\vec{F}$ in clockwise rotation $\vec{F}$ and $\vec{r}$ in the plane of paper.

Figures 4.6(a), 4.6(b) and 4.6(c) illustrate the directions involved. Figure 4.6(a) is a 3D drawing indicating the laminar (plane or two dimensional) object rotating about a (*fixed*) axis of rotation AOB, the axis being perpendicular to the object and passing through it. Figure 4.6(b) indicates the top view of the object when the rotation is in anticlockwise direction and Fig. 4.6(c) shows the view from the top, if rotation is in clockwise direction. (In fact, Figs. 4.6(b) and 4.6(c) are drawn in such a way that the applied force $\vec{F}$ and position vector $\vec{r}$ of the point of application of the force are in the plane of these figures). Direction of the torque is always perpendicular to the plane containing the vectors $\vec{r}$ and $\vec{F}$ and can be obtained from the rule of cross product or by using the right-hand thumb rule. In Fig. 4.6(b), it is perpendicular to the plane of the figure (in this case, perpendicular to the body) and outwards, i.e., coming out of the paper while in the Fig. 4.6(c), it is inwards, i.e., going into the paper.

In order to indicate the directions which are not in the plane of figure, we use a special convention: ⊙ for perpendicular to the plane of figure and outwards and ⊗ for perpendicular to the plane of figure and inwards.

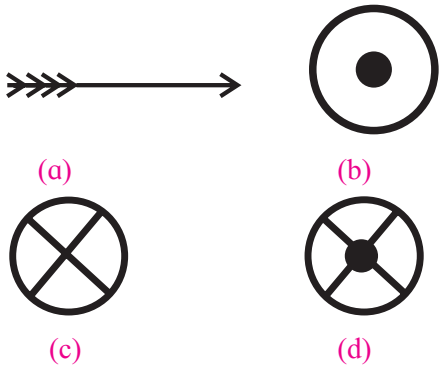


Fig. 4.7: Convention of pictorial representation of vectors as shown in (a) acting in a direction perpendicular to the plane of paper (b) coming out of paper, (c) going in to the paper and (d) perpendicular to the plane of paper.

This convention depends upon a traditional arrow shown in Fig. 4.7 (a). Consider yourself, looking towards the figure from the top. If this arrow approaches you, the tip of the arrow will be prominently seen. Hence circle with a

dot in it [Fig. 4.7 (b)] refers to perpendicular and outwards (or towards you). When you are leaving an arrow, i.e., if an arrow is going away from you, the feathers like a cross will be seen. Hence, a circle with a cross [Fig. 4.6 (c)] indicates perpendicular and inwards (or away from you). Circle with cross and dot indicates a line perpendicular to the plane of figure [Fig. 4.6 (d)].

Magnitude of torque, $\tau = r F \sin\theta$ --- (4.18) where θ is the smaller angle between the *directions* of $\vec{r}$ and $\vec{F}$.

Consequences: (i) If r or F is greater, the torque (hence the rotational effect) is greater. Thus, it is recommended to apply the force away from the hinges.

(ii) If $\theta = 90^0$, $\tau = \tau_{max} = rF$. Thus, the force should be applied along normal direction for easy rotation.

(iii) If $\theta = 0°$ or $180°$, $\tau = \tau_{min} = 0$. Thus, if the force is applied parallel or anti-parallel to $\vec{r}$, there is no rotation.

(iv) Moment of a force depends not only on the magnitude and direction of the force, but also on the point where the force acts with respect to the axis of rotation. Same force can have different torque as per its point of application.

4.11. Couple and its torque:

In the discussion of the torque given above, we had considered rotation of the body about a fixed axis and due to a single force. In real life, quite often we apply two equal and opposite forces acting along different lines of action in order to cause rotation. Common illustrations are turning a bicycle handle, turning the steering wheel, opening a common water tap, opening the lid of a bottle (rotation type), etc. Such a pair of forces consisting of two forces of equal magnitude acting in opposite directions along different lines of action is called a couple. It is used to realise a *purely* rotational motion. Moment of a couple or rotational effect of a couple is also called a *torque*.

It may be noted that in the discussion of rotation of a body about a fixed axis due to a single force, there is a reaction force at the fixed axis. Hence, for rotation one always needs two forces acting in opposite direction along different lines of action.

Torque or Moment of a couple: Figure 4.8 shows a couple consisting of two forces $\vec{F}_1$ *and* $\vec{F}_2$ of equal magnitudes and opposite directions acting along different lines of action separated by a distance r. Corresponding position vectors should now be defined with reference to the lines of action of forces. Position vector of *any* point on the line of action of force $\vec{F}_1$ from the line of action of force $\vec{F}_2$ is $\vec{r}_{12}$. Similarly, the position vector of *any* point on the line of action of force $\vec{F}_2$ from the line of action of force $\vec{F}_1$ is $\vec{r}_{21}$. Torque or moment of the couple is then given mathematically as

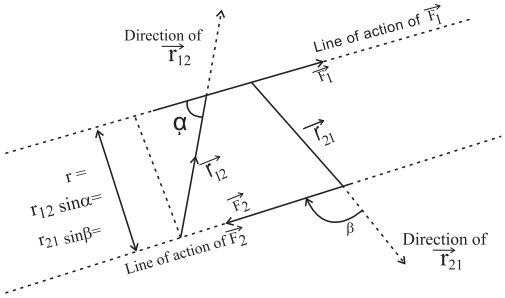


Fig. 4.8: Torque of a couple.

$$\vec{\tau} = \vec{r}_{12} \times \vec{F}_1 = \vec{r}_{21} \times \vec{F}_2 \qquad \text{--- (4.19)}$$

From the figure, it is clear that $r_{12} \sin\alpha = r_{21} \sin\beta = r$

If $|\vec{F}_1| = |\vec{F}_2| = F$, the magnitude of torque is given by

$$\tau = r_{12} F_1 \sin\alpha = r_{21} F_2 \sin\beta = r F \qquad \text{--- (4.20)}$$

It clearly shows that the torque corresponding to a given couple, i.e., the ***moment of a given couple is constant***, i.e., ***it is independent of the points of application of forces or the position of the axis of rotation***, but depends only upon magnitude of either force and the separation between their lines of action.

The direction of moment of couple can be obtained by using the vector formula of the torque or by using the right-hand thumb rule. For the couple shown in the Fig. 4.8, it is perpendicular to the plane of the figure and inwards. For a given pair of forces, the direction of the torque is fixed.

In many situations the word *couple* is used synonymous to moment of the couple or its torque, i.e., every time we may not say it as torque due to the couple, but say that a *couple* is acting.

	Moment of a force	Moment of a couple
1	$\vec{\tau} = \vec{r} \times \vec{f}$	$\vec{\tau} = \vec{r}_{12} \times \vec{F}_1 = \vec{r}_{21} \times \vec{F}_2$
2	$\vec{\tau}$ depends upon the axis of rotation and the point of application of the force.	$\vec{\tau}$ depends ***only*** upon the two forces, i.e., it is independent of the axis of rotation or the points of application of forces.
3	It can produce translational acceleration also, if the axis of rotation is not fixed or if friction is not enough.	Does not produce any translational acceleration, but produces only rotational or angular acceleration.
4	Its rotational effect can be balanced by a proper single force or by a proper couple.	Its rotational effect can be balanced *only* by another couple of equal and opposite torque.

4.11.1. To prove that the moment of a couple is independent of the axis of rotation:

Figure 4.9 shows a rectangular sheet (any object would do) free to rotate only about a fixed axis of rotation, perpendicular to the plane of figure, as shown. A couple consisting of forces $\vec{F}$ and $-\vec{F}$ is acting on the sheet at different locations.

Here we are considering the torque of a couple to be two torques due to individual forces causing rotation about the axis of rotation. In Fig. 4.9(a), the axis of rotation is between the lines of action of the two forces constituting the couple. Perpendicular distances of the axis of rotation from the forces $\vec{F}$ and $-\vec{F}$ are x and y respectively. Rotation due to the pair of forces *in this case* is anticlockwise (from top view), i.e., directions of individual torques due to the two forces are the same.

$$\therefore \tau = \tau_+ + \tau_- = xF + yF = (x+y)F = rF \quad (4.21)$$

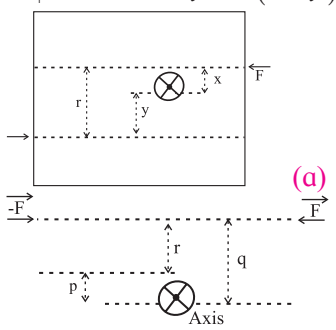


Fig. 4.9: Same couple on same object with fixed axis of rotation at different locations in (a) and (b).

In the Fig. 4.9 (b), lines of action of both the forces are on the same side of the axis of rotation. Thus, *in this case,* the rotation of $+\vec{F}$ is anticlockwise, while that of $-\vec{F}$ is clockwise (from the top view). As a result, their individual torques are oppositely directed. Perpendicular distance of the forces F and $-F$ from the axis of rotation are q and p respectively.

$$\therefore \tau = \tau_+ - \tau_- = qF - pF$$

$$= (q-p)F = rF \quad \text{--- (4.22)}$$

From equations (4.21) and (4.22), it is clear that the torque of a couple is independent of the axis of rotation.

4.12. Mechanical equilibrium:

As a consequence of Newton's second law, the momentum of a system is constant in the absence of an external unbalanced force. This state is called *mechanical equilibrium.*

A particle is said to be in mechanical equilibrium, if no net force is acting upon it. For a system of bodies to be in mechanical equilibrium, the net force acting on *any part* of the system should be zero. In other words, velocity or linear momentum of all parts of the system must be constant or (zero) for the system to be in mechanical equilibrium. Also, there is no acceleration in any part of the system.

Mathematically, $\sum \vec{F} = 0$, for any part of the system for mechanical equilibrium.

4.12.1 Stable, unstable and neutral equilibrium:

Figures 4.10 (a), (b) and (c) show a ball at rest in three situations under the action of balanced forces. In all these cases, it is under equilibrium. However, potential energy-wise, the three cases differ.

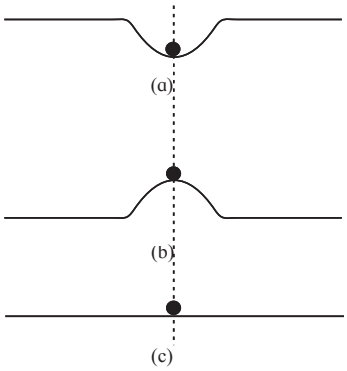


Fig. 4.10: states of mechanical equilibrium (a) stable, (b) unstable and (c) neutral.

Stable equilibrium: In Fig. 4.9(a), the ball is most stable and is said to be in *stable* equilibrium. If it is disturbed slightly from its equilibrium position and released, it tends to recover its position. In this case, potential energy of the system is at its local minimum.

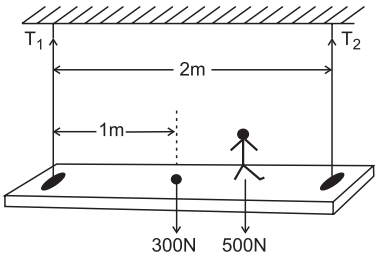
Unstable equilibrium: In Fig. 4.9(b), the ball is said to be in *unstable* equilibrium. If it is slightly disturbed from its equilibrium position, it moves farther from that position. This happens because initially, potential energy of the system is at its local maximum. If disturbed, it tries to achieve the configuration of minimum potential energy.

If potential energy function is known for the system, mathematically, the three equilibria can be explained with the help of derivatives of that function. At any equilibrium position, the first derivative of the potential energy function is zero $\left(\frac{dU}{dx}=0\right)$.

The sign of the second derivative $\left(\frac{d^2U}{dx^2}\right)$ decides the type of equilibrium. It is positive at stable equilibrium (or vice versa), negative at unstable equilibrium and zero (or does not exist) at neutral equilibrium configuration.

Neutral equilibrium: In Fig. 4.9(c), potential energy of the system is constant over a plane and remains same at any position. Thus, even if the ball is disturbed, it still remains in equilibrium at practically any position. This is described as *neutral* equilibrium.

Example 4.10: A uniform wooden plank of mass 30 kg is supported symmetrically by two light identical cables; each can sustain a tension up to 500 N. After tying, the cables are exactly vertical and are separated by 2 m. A boy of mass 50 kg, standing at the centre of the plank, is interested in walking on the plank. How far can he walk? ($g = 10$ ms^{-2})



Solution: Let T_1 and T_2 be the tensions along the cables, both acting vertically upwards.

Weight of the plank 300 N is acting vertically downwards through the centre, 1 m from either cable. Weight of the boy, 500 N is vertically downwards at the point where he is standing.

$\therefore T_1 + T_2 = 300 + 500 = 800$ N

Suppose that the boy is able to walk x m towards the right. Obviously, the tension in the right side cable goes on increasing as he walks towards the cable.

Moments of 300 N and 500 N forces about left end A are clockwise, while that of T_2 is anticlockwise.

As the cable can sustain 500 N, $(T_2)_{max}$ = 500 N

Thus, for the equilibrium about A, we can write,

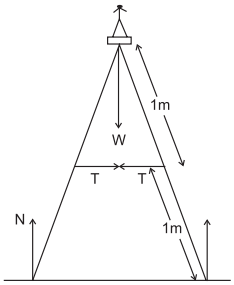
$300\times1+500\times(1+x)=500\times2 \quad \therefore x = 0.4$ m

Thus, the boy can walk up to 40 cm on either side of the centre.

Example 4.11: A ladder of negligible mass having a cross bar is resting on a frictionless horizontal floor with angle between its legs to be 40^0. Each leg is 1 m long. Calculate the

force experienced by the cross bar when a person of mass 50 kg is standing on the ladder. ($g = 10\ m\ s^{-2}$)

Solution: Tension T along the cross bar is horizontal. Let L be the length of each leg, which is 1 m.



As there is no friction, there is no horizontal reaction at the floor. Reaction N given by the floor at the base of the ladder will then be only vertical. Thus, along the vertical, two such reactions balance weight $W = mg$ of the person.

$$\therefore N = \frac{mg}{2} = 250\ \text{N}$$

At the left leg, about the upper end, the torque due to N is clockwise and that due to the tension T is anticlockwise. For equilibrium, these two torques should have same magnitude.

$$\therefore N.L\sin 20^0 = T.\left(\frac{L}{2}\right)\cos 20^0$$

$$\therefore T = 2N\ \tan 20^0 = 2\times 250\times 0.364 = 182\ \text{N}$$

4.13. Centre of mass:

As discussed earlier, Newton's laws of motion and many other laws are applicable for point masses only. However, in real life, we always come across finite objects (objects of measurable sizes). Concept of centre of mass (c.m.) helps us in considering these objects to be point objects at a particular location, thereby allowing us to apply Newton's laws of motion.

4.13.1. Mathematical understanding of centre of mass:

(i) System of n particles: Consider a system of n particles of masses $m_1, m_2 \ldots m_n$.

$$\therefore \sum_1^n m_i = M \text{ the total mass.}$$

Let $\vec{r}_1, \vec{r}_2, \ldots \vec{r}_n$ be their respective position vectors from a given origin O (Fig. 4.11) .

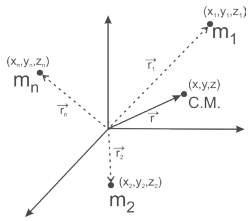


Fig. 4.11: Centre of mass for n particles.

Position vector $\vec{r}$ of their centre of mass from the same origin is then given by

$$\vec{r} = \frac{\sum_1^n m_i\vec{r}_i}{\sum_1^n m_i} = \frac{\sum_1^n m_i\vec{r}_i}{M}$$

If the origin itself is at the centre of mass,

$\vec{r} = 0\ \therefore \sum_1^n m_i\vec{r}_i = 0$, then

$\sum_1^n m_i\vec{r}_i$ gives the moment of masses (similar to moment of force) about the centre of mass.

Thus, centre of mass is a point about which the summation of moments of masses in the system is zero.

If $x_1, x_2, \ldots x_n$ are the respective x-coordinates of $r_1, r_2, \ldots r_n$, the x-coordinate of the centre of mass is given by

$$x = \frac{\sum_1^n m_i x_i}{\sum_1^n m_i} = \frac{\sum_1^n m_i x_i}{M}$$

Similarly, y and z-coordinates of the centre of mass are respectively given by

$$y = \frac{\sum_1^n m_i y_i}{\sum_1^n m_i} = \frac{\sum_1^n m_i y_i}{M}$$

and

$$z = \frac{\sum_1^n m_i z_i}{\sum_1^n m_i} = \frac{\sum_1^n m_i z_i}{M}$$

(i) Continuous mass distribution: For a continuous mass distribution ***with uniform density***, we need to use integration instead of summation. In this case, the position vector of the centre of mass is given by

$$\vec{r} = \frac{\int \vec{r}\ dm}{\int dm} = \frac{\int \vec{r}\ dm}{M},$$

where $\int dm = M$ is the total mass of the object. Then the Cartesian coordinates of c.m. are

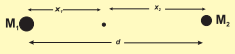
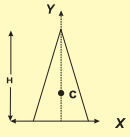
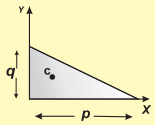
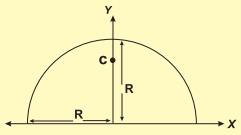
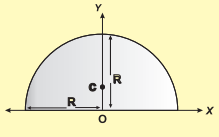
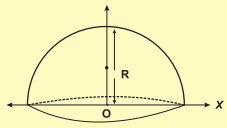
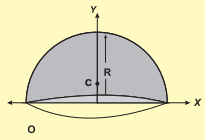
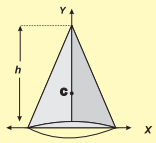
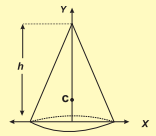
$$x = \frac{\int x\,dm}{\int dm} = \frac{\int x\,dm}{M}$$

$$y = \frac{\int y\,dm}{\int dm} = \frac{\int y\,dm}{M}$$

$$z = \frac{\int z\,dm}{\int dm} = \frac{\int z\,dm}{M}$$

Using the expressions given above, the centres of mass of uniform symmetric objects can be obtained. Some of these are listed in the Table 4.1 given below:

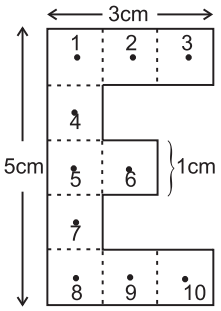
Table 4.1: Coordinate of the centre of mass (c.m.) for some symmetrical objects

Coordinates of c.m.	Uniform Symmetric Objects
System of two point masses: c.m. divides the distance in inverse proportion of the masses	
Any geometrically symmetric object of uniform density.	Centre of mass at a geometrical centre of the object
Isosceles triangular plate $x_c = 0, y_c = \frac{H}{3}$	
Right angled triangular plate $x_c = \frac{p}{3}, y_c = \frac{q}{3}$	
Thin semicircular ring of radius R $x_c = 0, y_c = \frac{2R}{\pi}$	
Thin semicircular disc of radius R $x_c = 0, y_c = \frac{4R}{3\pi}$	
Hemispherical shell of radius R $x_c = 0, y_c = \frac{R}{2}$	
Solid hemisphere of radius R $x_c = 0, y_c = \frac{3R}{8}$	
Hollow right circular cone of height h $x_c = 0, y_c = \frac{h}{3}$	
Solid right circular cone of height h $x_c = 0, y_c = \frac{h}{4}$	

Example 4.12: A letter 'E' is prepared from a uniform cardboard with shape and dimensions as shown in the figure. Locate its centre of mass.

Solution: As the sheet is uniform, each square can be taken to be equivalent to mass m concentrated at its respective centre. These masses will then be at the points labelled with numbers 1 to 10, as shown in figure. Let us select the origin to be at the left central mass m_5, as shown and all the co-ordinates to be in cm.

By symmetry, the centre of mass of m_1, m_2 and m_3 will be at $m_2(1, 2)$ having effective mass $3m$. Similarly, effective mass $3m$ due to m_8, m_9 and m_{10} will be at $m_9(1, -2)$. Again, by symmetry, the centre of mass of these two ($3m$ each) will have co-ordinates (1, 0). Mass m_6 is also having co-ordinates (1, 0). Thus, the effective mass at (1, 0) is $7m$.



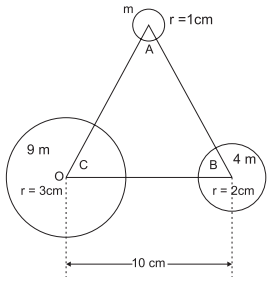
Using symmetry for m_4, m_5 and m_7, there will be effective mass $3m$ at the origin (0, 0).

Thus, effectively, $3m$ and $7m$ are separated by 1 cm along x-direction. y-coordinate is not required.

$$x_c = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} = \frac{3\times 0 + 7\times 1}{3+7} = 0.7\,\text{cm}$$

Alternately, for two point masses, the centre of mass divides the distance between them in the inverse ratio of their masses. Hence, 1 cm is divided in the ratio 7:3. $\therefore x_c = \frac{7}{7+3}\times 1 = 0.7\,\text{cm}$ from $3m$, i.e., from the origin at m_c

Example 4.13: Three thin walled uniform hollow spheres of radii 1cm, 2 cm and 3 cm are so located that their centres are on the three vertices of an equilateral triangle ABC having each side 10 cm. Determine centre of mass of the system.



Solution: Mass of a thin walled uniform hollow sphere is proportional to its surface area, (as density is constant) hence proportional to r^2. Thus, if mass of the sphere at A is $m_A = m$, then $m_B = 4m$ and $m_C = 9m$. By symmetry of the spherical surface, their centres of mass are at their respective centres, i.e., at A, B and C.

Let us choose the origin to be at C, where the largest mass $9m$ is located and the point B with mass $4m$ on the positive x-axis. With this, the co-ordinates of C are (0, 0) and that of B are (10, 0). (Locating the origin at the larger mass here save our efforts of calculations like multiplications with larger numbers). If A of mass m is taken in the first quadrant, its co-ordinates will be $\left[5,\ \frac{10\sqrt{3}}{2}\right]$

$$x_c = \frac{m_Ax_A + m_Bx_B + m_Cx_C}{m_A + m_B + m_C}$$

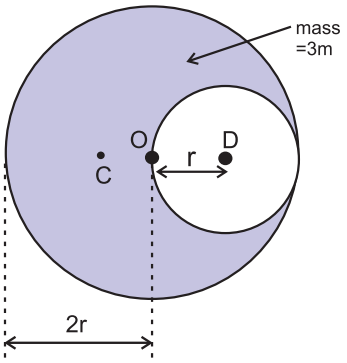
$$= \frac{m\times 5 + 4m\times 10 + 9m\times 0}{m+4m+9m} = \frac{45}{14}\,\text{cm}$$

$$y_c = \frac{m_Ay_A + m_By_B + m_Cy_C}{m_A + m_B + m_C}$$

$$= \frac{m\times\frac{10\sqrt{3}}{2} + 4m\times 0 + 9m\times 0}{m+4m+9m} = \frac{10\sqrt{3}}{28}\,\text{cm}$$

Example 4.14: A hole of radius r is cut from a uniform disc of radius $2r$. Centre of the hole is at a distance r from centre of the disc. Locate centre of mass of the remaining part of the disc.

Solution: **Method I:** (Using entire disc): Before cutting the hole, c.m. of the full disc was at its centre. Let this be our origin O. Centre of mass of the cut portion is at its centre D. Thus, it is at a distance $x_1 = r$ form the origin. Let C be the centre of mass of the remaining disc. Obviously, it should be on the extension of the line DO. Let it be at a distance $x_2 = x$ from the origin. As the disc is uniform, mass of any of its part is proportional to the area of that part.



Thus, if m is the mass of the cut disc, mass of the entire disc must be $4m$ and mass of the remaining disc will be $3m$.

$$x_c = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$ As centre of mass of the full disc is at the origin, we can write,

$$0 = \frac{m\times r + 3m\times(x)}{m+3m} \quad \therefore x = \frac{-r}{3}$$

Method II: (Using negative mass): Let $\vec{R}$ be the position vector of the centre of mass of the uniform disc of mass M. Mass m is with centre of mass at position vector $\vec{r}$ from the centre of the disc. Position vector of the centre of mass of the remaining disc is then given by

$\vec{r}_c = \frac{M\vec{R} - m\vec{r}}{M - m}$ ..... (as if there is a negative mass, i.e., $m_2 = -m$)

With our description, $M = 4m$, $m = m$, $R = 0$ and $r = r$ $\therefore r_c = \frac{-mr}{3m} = \frac{-r}{3}$ ... Same as method I.

4.13.2. Velocity of centre of mass:

Let $v_1, v_2, \ldots v_n$ be the velocities of a system of point masses $m_1, m_2, \ldots m_n$. Velocity of the centre of mass of the system is given by

$$\vec{v}_{cm} = \frac{\sum_1^n m_i \vec{v}_i}{\sum_1^n m_i} = \frac{\sum_1^n m_i \vec{v}_i}{M}$$

$$= \frac{\text{resultant linear momentum}}{\text{total mass}}$$

= weighted average of momenta

x, y and z components of $\vec{v}$ can be obtained similarly.

For continuous distribution, $\vec{v}_{cm} = \frac{\int \vec{v}\, dm}{M}$

4.13.3. Acceleration of the centre of mass:

Let $a_1, a_2, \ldots a_n$ be the accelerations of a system of point masses $m_1, m_2, \ldots m_n$. Acceleration of the centre of mass of the system is given by

$$\vec{a}_{cm} = \frac{\sum_1^n m_i \vec{a}_i}{\sum_1^n m_i} = \frac{\sum_1^n m_i \vec{a}_i}{M}$$

$$= \frac{\text{resultant force}}{\text{total mass}}$$

= weighted average of forces

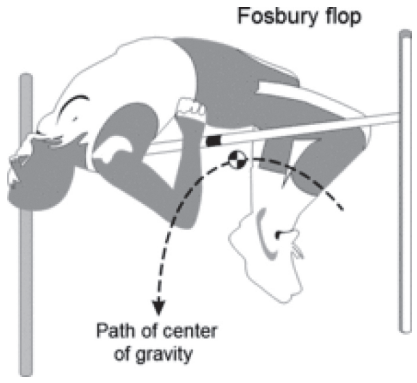
x, y and z components of $\vec{a}$ can be obtained similarly.

For continuous distribution, $\vec{a}_{cm} = \frac{\int \vec{a}\, dm}{M}$

4.13.4. Characteristics of centre of mass:

1. Centre of mass is a hypothetical point at which entire mass of the body can be assumed to be concentrated.
2. Centre of mass is a location, and not a physical quantity.
3. Centre of mass is particle equivalent of a given object for applying laws of motion.
4. Centre of mass is the point at which, if a force is applied, it causes only linear acceleration and not angular acceleration.
5. Centre of mass is located at the centroid, for a rigid body ***of uniform density.***
6. Centre of mass is located at the geometrical centre, for a symmetric rigid body ***of uniform density.***
7. Location of centre of mass can be changed ***only*** by an external unbalanced force.
8. Internal forces (like during collision or explosion) ***never*** change the location of centre of mass.
9. Position of the centre of mass depends only upon the distribution of mass, however, to describe its location we may use a coordinate system with a *suitable* origin. In statistical terms the centre of mass is decided by the *weighted* average of individual masses. This is obtained by giving proper mass *weightage* to the distance. This should be clear from the mathematical expression for the location of the centre of mass.
10. For a system of particles, the centre of mass *need not* coincide with any of the particles.
11. While balancing an object on a pivot, the line of action of *weight* must pass through the centre of mass and the pivot. Quite often, this is an unstable equilibrium.
12. Centre of mass of a system of only two particles divides the distance between the particles in an inverse ratio of their masses, i.e., it is closer to the heavier mass.
13. Centre of mass is a point about which the summation of moments of masses in the system is zero.
14. If there is an axial symmetry for a given object, the centre of mass lies on the axis of symmetry.
15. If there are multiple axes of symmetry for a given object, the centre of mass is at their point of intersection.
16. Centre of mass need not be within the

body (See the photograph given below: Picture 4.1). Other examples are a ring, a horse shoe, etc.



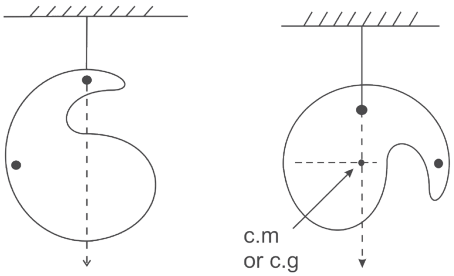
Picture 4.1: Courtesy Wikipedia: Estimated center of mass/gravity of a high jumper doing a Fosbury Flop. Note that it is below the bar in this position. This is possible because our head and legs are much heavier than the fleshy part. Increase in the gravitational potential energy of the high jumper depends upon this point.

4.14. Centre of gravity

Centre of gravity (c.g.) of a body is the point around which the resultant torque due to force of gravity on the body is zero. Analogous to centre of mass, it is the weighted average of the gravitational forces (weights) on individual particles.

For uniform gravitational field (in simple words, if g is constant), c.g. always coincides with the c.m. Obviously it is true for all the objects *on the Earth* in our daily life. Thus, in common usage, the terms c.g. and c.m. are used for same purpose. This property can be used to determine the c.g. (or c.m.) of a laminar (laminar means like a leaf – two dimensional) object.

In Fig. 4.12, a laminar object is suspended from a rigid support at two orientations. Lines are to be drawn on the object parallel to the plumb line shown. Plumb line is always vertical, i.e., parallel to the line of action of gravitational force. Intersection of the lines drawn is then the point through which line of action of the gravitational force passes for any orientation. Thus, it gives the location of the c.g. or c.m.



Centre of mass is a fixed property for a given rigid body in spite of any orientation. The centre of gravity may depend upon non-uniformity of the gravitational field, in turn, will depend upon the orientation. For objects on the Earth , this will be possible only if the size of an object is comparable to that of the Earth (size at least few thousand km). In such cases, the c.g. will be slightly lower than the c.m. as on the lower side of an object the gravitational field is stronger. Of course, we shall not come across such an object.

Exercises

1. Choose the correct answer.

i) Consider following pair of forces of equal magnitude and opposite directions:
(P) Gravitational forces exerted on each other by two point masses separated by a distance.
(Q) Couple of forces used to rotate a water tap.
(R) Gravitational force and normal force experienced by an object kept on a table.
For which of these pair/pairs the two forces do NOT cancel each other's translational effect?
(A) Only P (B) Only P and Q
(C) Only R (D) Only Q and R

ii) Consider following forces: (w) Force due to tension along a string, (x) Normal force given by a surface, (y) Force due to air resistance and (z) Buoyant force or upthrust given by a fluid.
Which of these are electromagnetic forces?
(A) Only w, y and z

(B) Only w, x and y
(C) Only x, y and z
(D) All four.

iii) At a given instant three point masses m, $2m$ and $3m$ are equidistant from each other. Consider only the gravitational forces between them. Select correct statement/s for this instance only:
(A) Mass m experiences maximum force.
(B) Mass $2m$ experiences maximum force.
(C) Mass $3m$ experiences maximum force.
(D) All masses experience force of same magnitude.

iv) The rough surface of a horizontal table offers a definite *maximum* opposing force to initiate the motion of a block along the table, which is proportional to the resultant normal force given by the table. Forces F_1 and F_2 act at the same angle θ with the horizontal and both are just initiating the sliding motion of the block along the table. Force F_1 is a pulling force while the force F_2 is a pushing force. $F_2 > F_1$, because
(A) Component of F_2 adds up to weight to increase the normal reaction.
(B) Component of F_1 adds up to weight to increase the normal reaction.
(C) Component of F_2 adds up to the opposing force.
(D) Component of F_1 adds up to the opposing force.

v. A mass $2m$ moving with some speed is *directly* approaching another mass m moving with double speed. After some time, they collide with coefficient of restitution 0.5. Ratio of their respective speeds after collision is
(A) 2/3 (B) 3/2
(C) 2 (D) ½

vi. A uniform rod of mass $2m$ is held horizontal by two sturdy, practically inextensible vertical strings tied at its ends. A boy of mass $3m$ hangs himself at one third length of the rod. Ratio of the tension in the string close to the boy to that in the other string is
(A) 2 (B) 1.5
(C) 4/3 (D) 5/3

vii. Select WRONG statement about centre of mass:
(A) Centre of mass of a 'C' shaped uniform rod can never be a point on that rod.
(B) If the line of action of a force passes through the centre of mass, the moment of that force is zero.
(C) Centre of mass of our Earth is not at its geometrical centre.
(D) While balancing an object on a pivot, the line of action of the gravitational force of the earth passes through the centre of mass of the object.

viii. For which of the following objects will the centre of mass NOT be at their geometrical centre?
(I) An egg
(II) a cylindrical box full of rice
(III) a cubical box containing assorted sweets
(A) Only (I)
(B) Only (I) and (II)
(C) Only (III)
(D) All, (I), (II) and (III).

2. Answer the following questions.

i) In the following table, every entry on the left column can match with any number of entries on the right side. Pick up ***all*** those and write respectively against A, B, C and D.

Name of the force		Type of the force	
A	Force due to tension in a string	P	EM force
B	Normal force	Q	Reaction force
C	Frictional force	R	Conservative force
D	Resistive force offered by air or water for objects moving through it.	S	Non-conservative force

ii) In real life objects, never travel with uniform velocity, even on a horizontal surface, unless *something* is done? Why is it so? What is to be done?

iii) For the study of any kind of motion, we never use Newton's first law of motion directly. Why should it be studied?

iv) Are there any situations in which we cannot apply Newton's laws of motion? Is there any alternative for it?

v) You are inside a closed capsule from where you are not able to see anything about the outside world. Suddenly you feel that you are pushed towards your right. Can you explain the possible cause (s)? Is it a feeling or a reality? Give at least one more situation like this.

vi) Among the four fundamental forces, only one force governs your daily life almost entirely. Justify the statement by stating that force.

vii) Find the odd man out: (i) Force responsible for a string to become taut on stretching (ii) Weight of an object (iii) The force due to which we can hold an object in hand.

viii) You are sitting next to your friend on ground. Is there any gravitational force of attraction between you two? If so, why are you not coming together naturally? Is any force other than the gravitational force of the earth coming in picture?

ix) Distinguish between: (A) Real and pseudo forces, (B) Conservative and non-conservative forces, (C) Contact and non-contact forces, (C) Inertial and non-inertial frames of reference.

x) State the formula for calculating work done by a force. Are there any conditions or limitations in using it directly? If so, state those clearly. Is there any mathematical way out for it? Explain.

xi) Justify the statement, "Work and energy are the two sides of a coin".

xii) From the terrace of a building of height H, you dropped a ball of mass m. It reached the ground with speed v. Is the relation $mgH = \frac{1}{2}mv^2$ applicable exactly? If not, how can you account for the difference? Will the ball bounce to the same height from where it was dropped?

xiii) State the law of conservation of linear momentum. It is a consequence of which law? Given an example from our daily life for conservation of momentum. Does it hold good during burst of a cracker?

xiv) Define coefficient of restitution and obtain its value for an elastic collision and a perfectly inelastic collision.

xv) Discuss the following as special cases of elastic collisions and obtain their exact or approximate final velocities in terms of their initial velocities.
(i) Colliding bodies are identical.
(ii) A veru heavy object collides on a lighter object, initially at rest.
(iii) A very light object collides on a comparatively much massive object, initially at rest.

xvi) A bullet of mass m_1 travelling with a velocity u strikes a stationary wooden block of mass m_2 and gets embedded into it. Determine the expression for loss in the kinetic energy of the system. Is this violating the principle of conservation of energy? If not, how can you account for this loss?

xvii) One of the effects of a force is to change the momentum. Define the quantity related to this and explain it for a variable force. Usually when do we define it instead of using the force?

xviii) While rotating an object or while opening a door or a water tap we apply a force or forces. Under which conditions is this process easy for us? Why? Define the vector quantity concerned. How does it differ for a single force and for two opposite forces with different lines of action?

xix) Why is the moment of a couple independent of the axis of rotation even if the axis is fixed?

xx) Explain balancing or mechanical equilibrium. Linear velocity of a rotating fan as a whole is generally zero. Is it in mechanical equilibrium? Justify your answer.

xxi) Why do we need to know the centre of mass of an object? For which objects, its position may differ from that of the centre of gravity?

Use g = 10* m s^{-2}, *unless, otherwise stated.

3. Solve the following problems.

i) A truck of mass 5 ton is travelling on a

horizontal road with 36 km hr $^{-1}$ stops on traveling 1 km after its engine fails suddenly. What fraction of its weight is the frictional force exerted by the road?
If we assume that the story repeats for a car of mass 1 ton i.e., can moving with same speed stops in similar distance same how much will the fraction be?

[Ans: $\frac{1}{200}$ in the both]

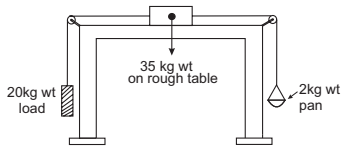
ii) A lighter object *A* and a heavier object *B* are initially at rest. Both are imparted the same linear momentum. Which will start with greater kinetic energy: *A* or *B* or both will start with the same energy?

[Ans: *A*]

iii) As I was standing on a weighing machine inside a lift it recorded 50 kg wt. Suddenly for few seconds it recorded 45 kg wt. What must have happened during that time? Explain with complete numerical analysis.
[Ans: Lift must be coming down with acceleration $\frac{g}{10} = 1\text{m s}^{-2}$]

iv) Figure below shows a block of mass 35 kg resting on a table. The table is so rough that it offers a self adjusting resistive force 10% of the weight of the block for its sliding motion along the table. A 20 kg wt load is attached to the block and is passed over a pulley to hang freely on the left side. On the right side there is a 2 kg wt pan attached to the block and hung freely. Weights of 1 kg wt each, can be added to the pan. Minimum how many and maximum how many such weights can be added into the pan so that the block does not slide along the table?

[Ans: Min 15, maximum 21].



v) Power is rate of doing work or the rate at which energy is supplied to the system. A constant force *F* is applied to a body of mass *m*. Power delivered by the force at time *t* from the start is proportional to
(a) *t* (b) t^2
(c) $\sqrt{t}$ (d) t^0
Derive the expression for power in terms of *F*, *m* and *t*.

[Ans: $p = \frac{F^2 t}{m}, \therefore p \alpha t$]

vi) 40000 litre of oil of density 0.9 g cc is pumped from an oil tanker ship into a storage tank at 10 m higher level than the ship in half an hour. What should be the power of the pump?

[Ans: 2 kW]

vii) Ten identical masses (*m* each) are connected one below the other with 10 strings. Holding the topmost string, the system is accelerated upwards with acceleration g/2. What is the tension in the 6th string from the top (Topmost string being the first string)?

[Ans: 6 *mg*]

viii) Two galaxies of masses 9 billion solar mass and 4 billion solar mass are 5 million light years apart. If, the Sun has to cross the line joining them, without being attracted by either of them, through what point it should pass?
[Ans: 3 million light years from the 9 billion solar mass]

ix) While decreasing linearly from 5 N to 3 N, a force displaces an object from 3 m to 5 m. Calculate the work done by this force during this displacement.

[Ans: 8 N]

x) Variation of a force in a certain region is given by $F = 6x^2 - 4x - 8$. It displaces an object from $x = 1$ m to $x = 2$ m in this region. Calculate the amount of work done. [Ans: Zero]

xi) A ball of mass 100 g dropped on the ground from 5 m bounces repeatedly. During every bounce 64% of the potential energy is converted into kinetic energy. Calculate the following:
(a) Coefficient of restitution.
(b) Speed with which the ball comes up from the ground after third bounce.
(c) Impulse given by the ball to the ground during this bounce.
(d) Average force exerted by the ground

if this impact lasts for 250 ms.

(e) Average pressure exerted by the ball on the ground during this impact if contact area of the ball is 0.5 cm^2.

[Ans: 0.8, 5.12 m/s, 1.152N s, 4.608 N, 9.216×10^4 N/m^2]

xii) A spring ball of mass 0.5 kg is dropped from some height. On falling freely for 10 s, it explodes into two fragments of mass ratio 1:2. The lighter fragment continues to travel downwards with speed of 60 m/s. Calculate the kinetic energy supplied during explosion.

[Ans: 200 J]

xiii) A marble of mass $2m$ travelling at 6 cm/s is directly followed by another marble of mass m with double speed. After collision, the heavier one travels with the average initial speed of the two. Calculate the coefficient of restitution.

[Ans: 0.5]

xiv) A, 2 m long wooden plank of mass 20 kg is pivoted (supported from below) at 0.5 m from either end. A person of mass 40 kg starts walking from one of these pivots to the farther end. How far can the person walk before the plank topples?

[Ans: 1.25 m]

xv) A 2 m long ladder of mass 10 kg is kept against a wall such that its base is 1.2 m away from the wall. The wall is smooth but the ground is rough. Roughness of the ground is such that it offers a maximum horizontal resistive force (for sliding motion) half that of normal reaction at the point of contact. A monkey of mass 20 kg starts climbing the ladder. How far can it climb ***along*** the ladder? How much is the horizontal reaction at the wall?

[Ans: 1.5 m, 15 N]

xvi) Four uniform solid cubes of edges 10 cm, 20 cm, 30 cm and 40 cm are kept on the ground, touching each other in order. Locate centre of mass of their system.

[Ans: 65 cm, 17.7 cm]

xvii) A uniform solid sphere of radius R has a hole of radius $R/2$ drilled inside it. One end of the hole is at the centre of the sphere while the other is at the boundary. Locate centre of mass of the remaining sphere.

[Ans: -R/14]

xviii) In the following table, every item on the left side can match with any number of items on the right hand side. Select all those.

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Types of collision	Illustrations
(a) Elastic collision (b) Inelastic collision (c) Perfectly inelastic collision (d) Head on collision	(i) A ball hit by a bat. (ii) Molecular collisions responsible for pressure exerted by a gas. (iii) A stationary marble A is hit by marble B and the marble B comes to rest. (iv) A blob of clay dropped on the ground sticks to the ground. (v) Out of anger, giving a kick to a wall. (vi) A striker hits the boundary of a carrom board in a direction perpendicular to the boundary and rebounds.